

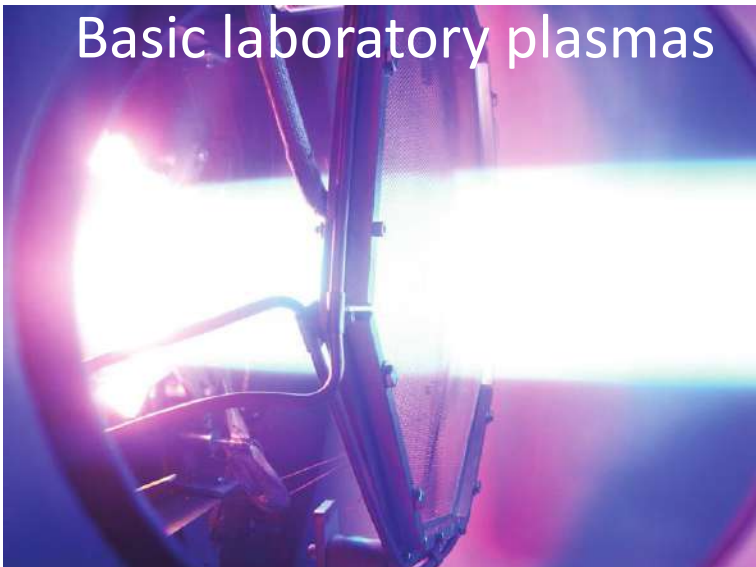
**Operation and
stability of an
emissive cathode
in a high density
plasma**

Laboratoire de
Physique à
l'ENS de Lyon

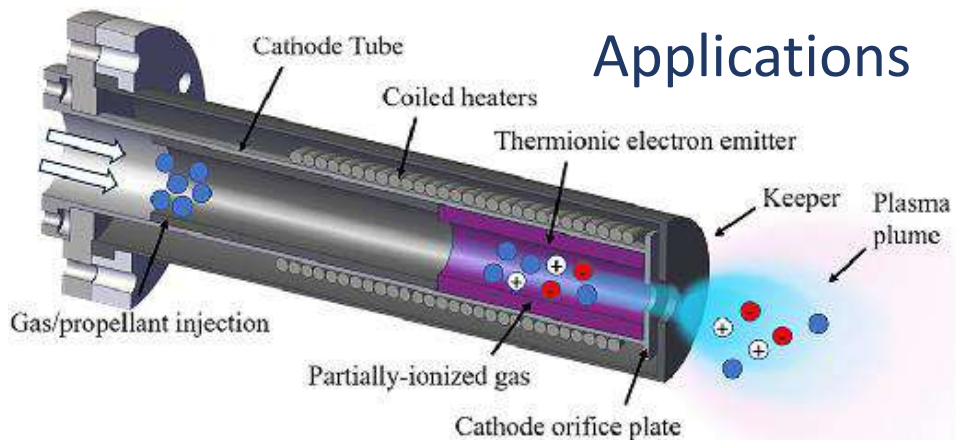
Francis Pagaud,
Vincent Dolique,
Nicolas Plihon

Plasma ignition

Basic laboratory plasmas

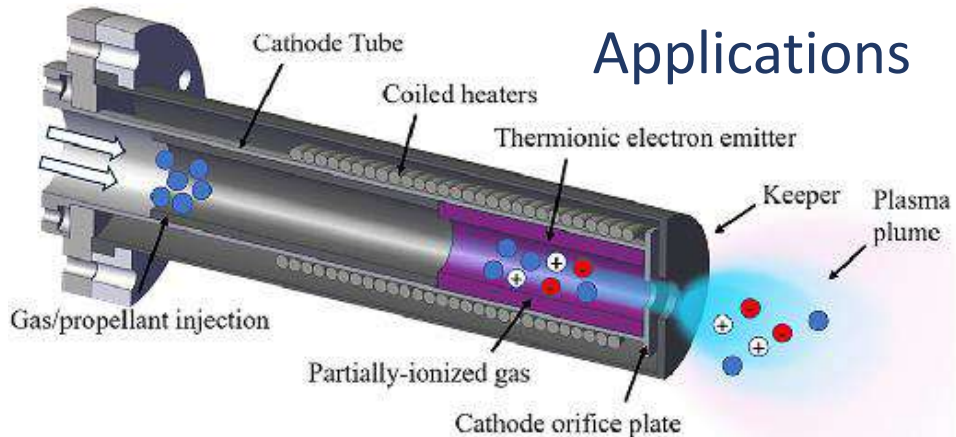
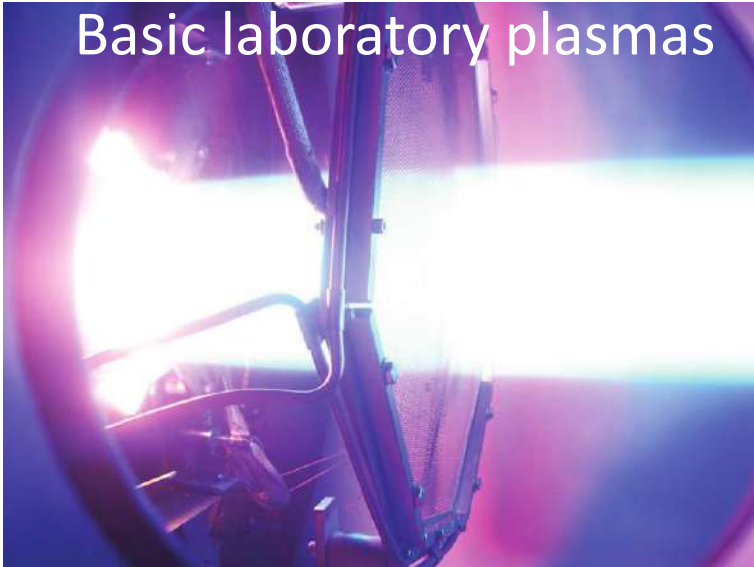


Applications



Plasma ignition

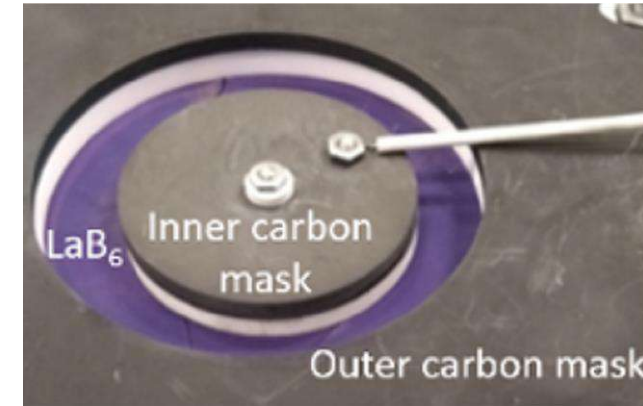
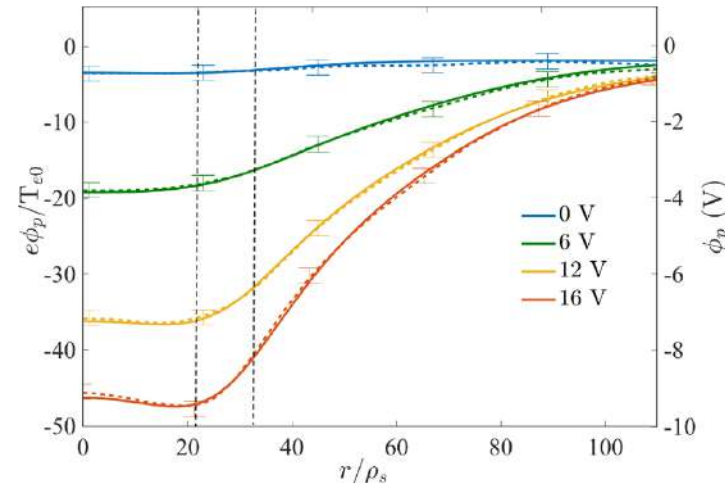
Basic laboratory plasmas



Applications

Plasma control

Plasma potential control, LAPD

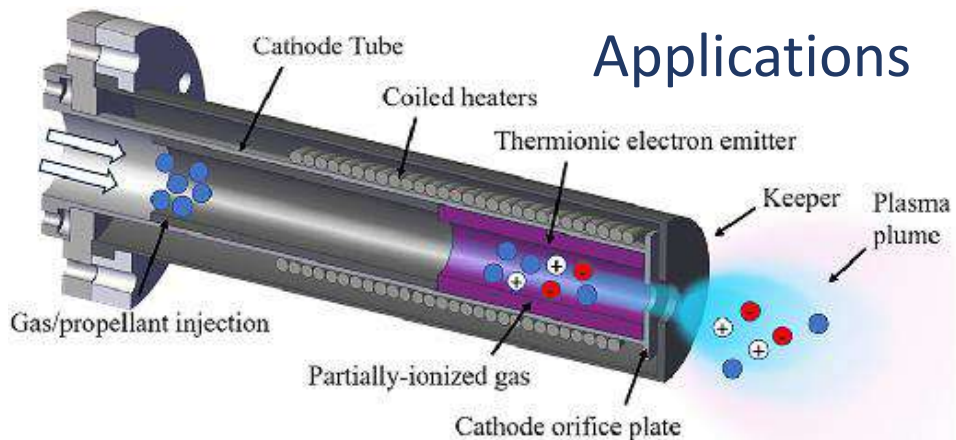
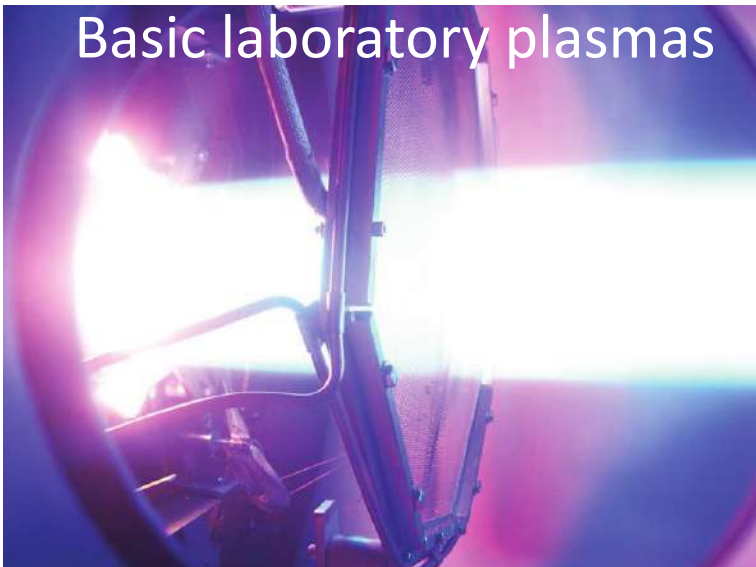


Emissive cathodes

4

Plasma ignition

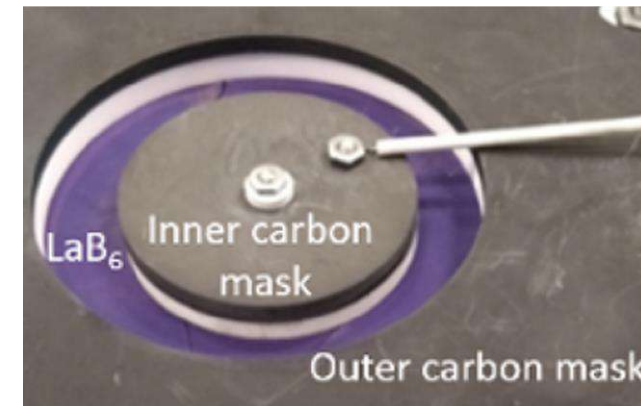
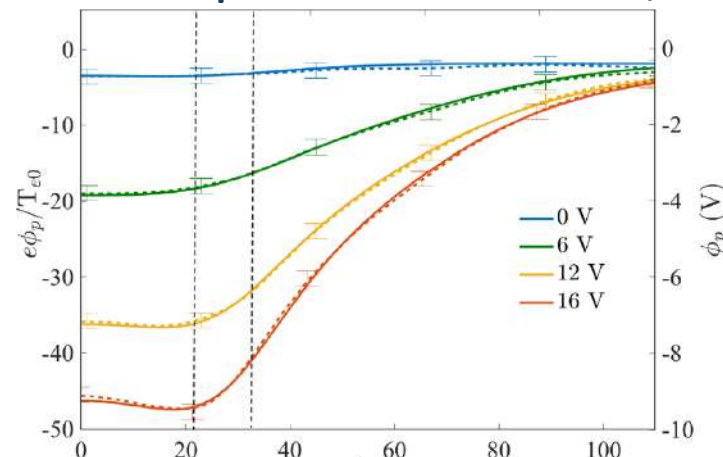
Basic laboratory plasmas



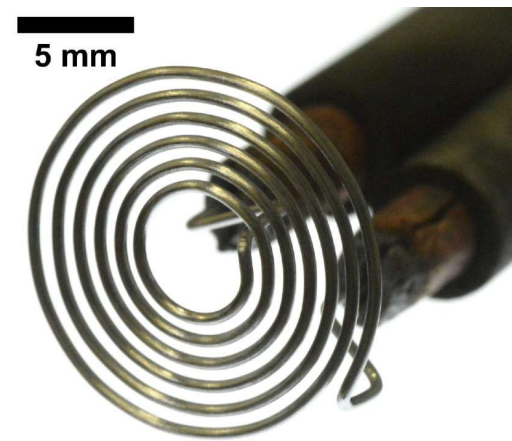
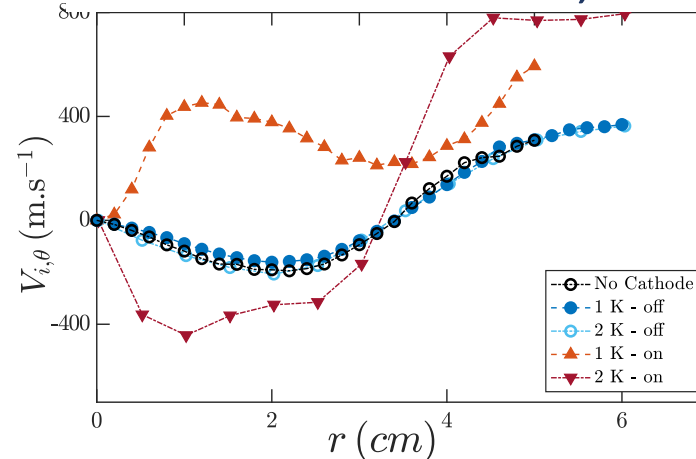
Applications

Plasma control

Plasma potential control, LAPD

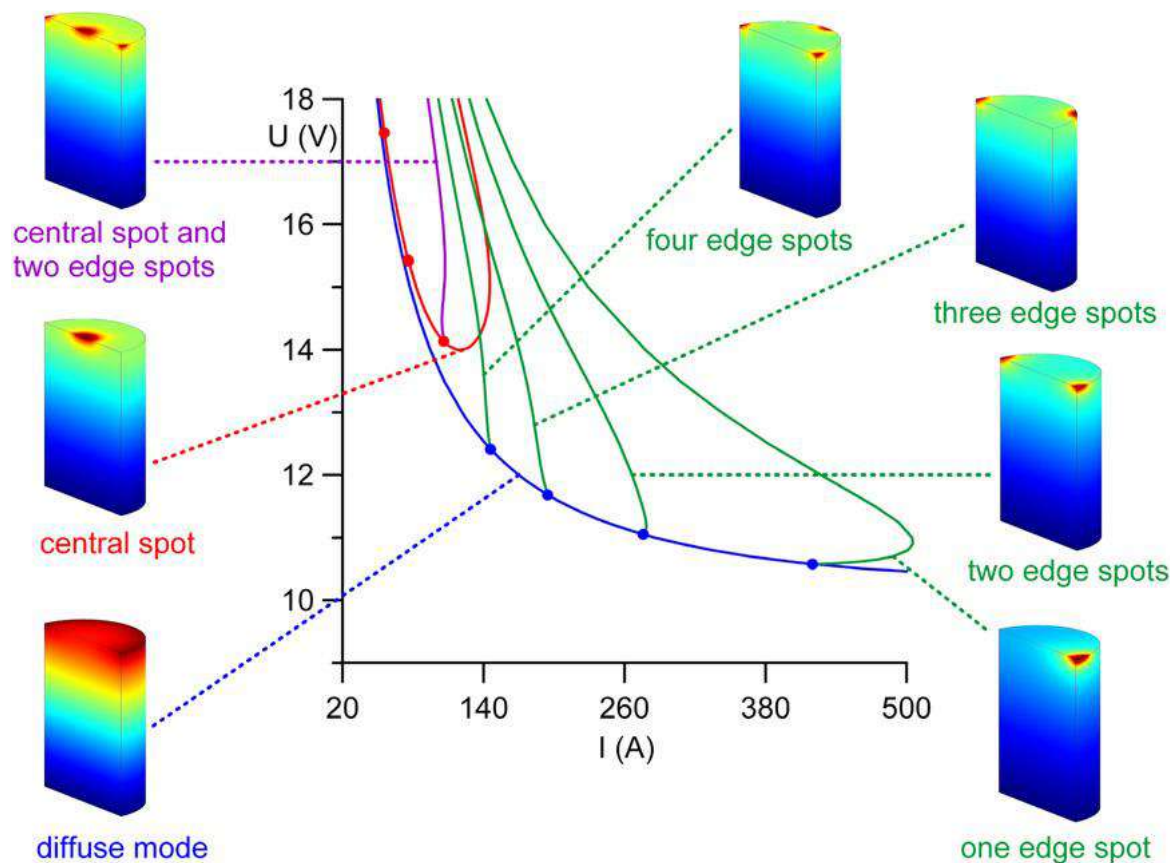


Plasma flow control, VKP



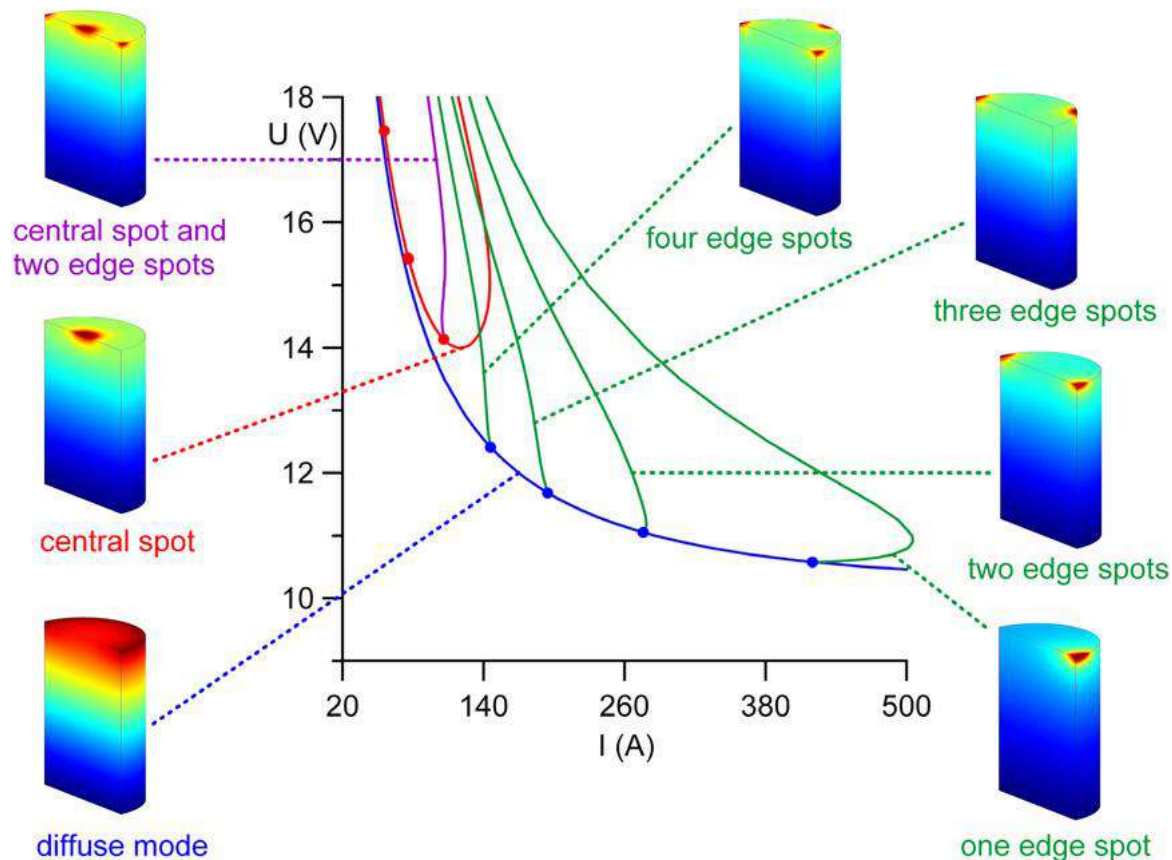
➤ Renewed interest

High-pressure arc discharges



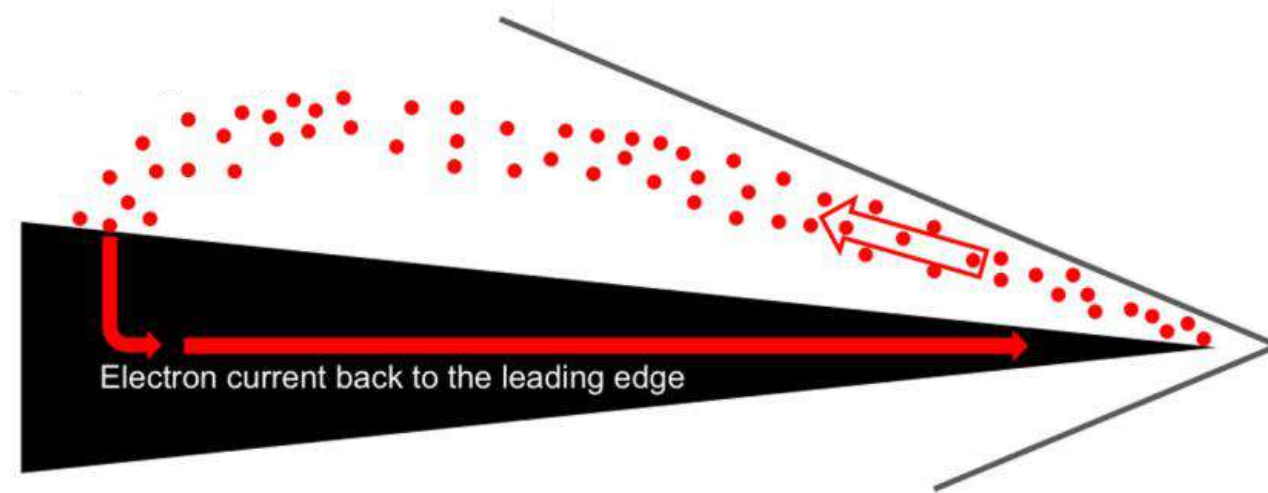
➤ Complex plasma-cathode interactions

High-pressure arc discharges



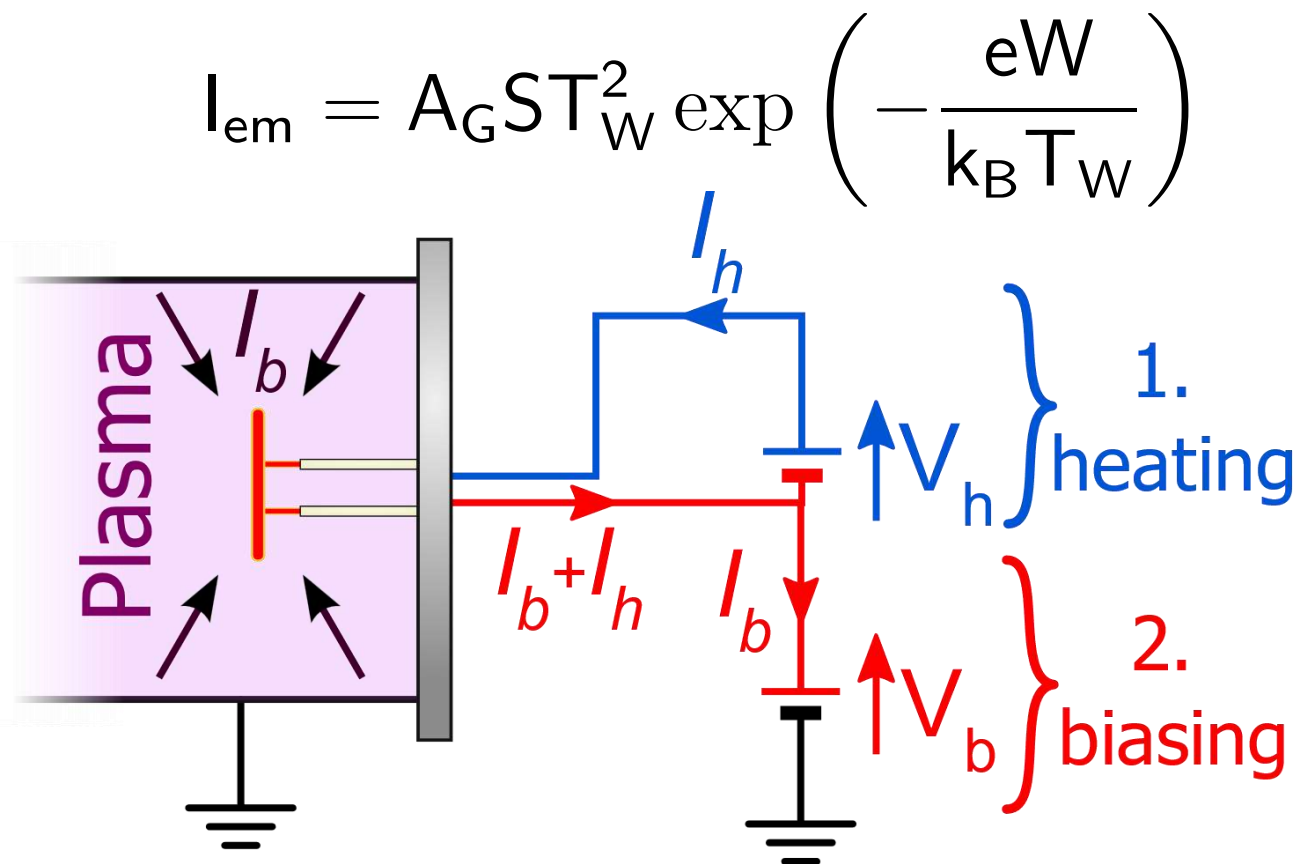
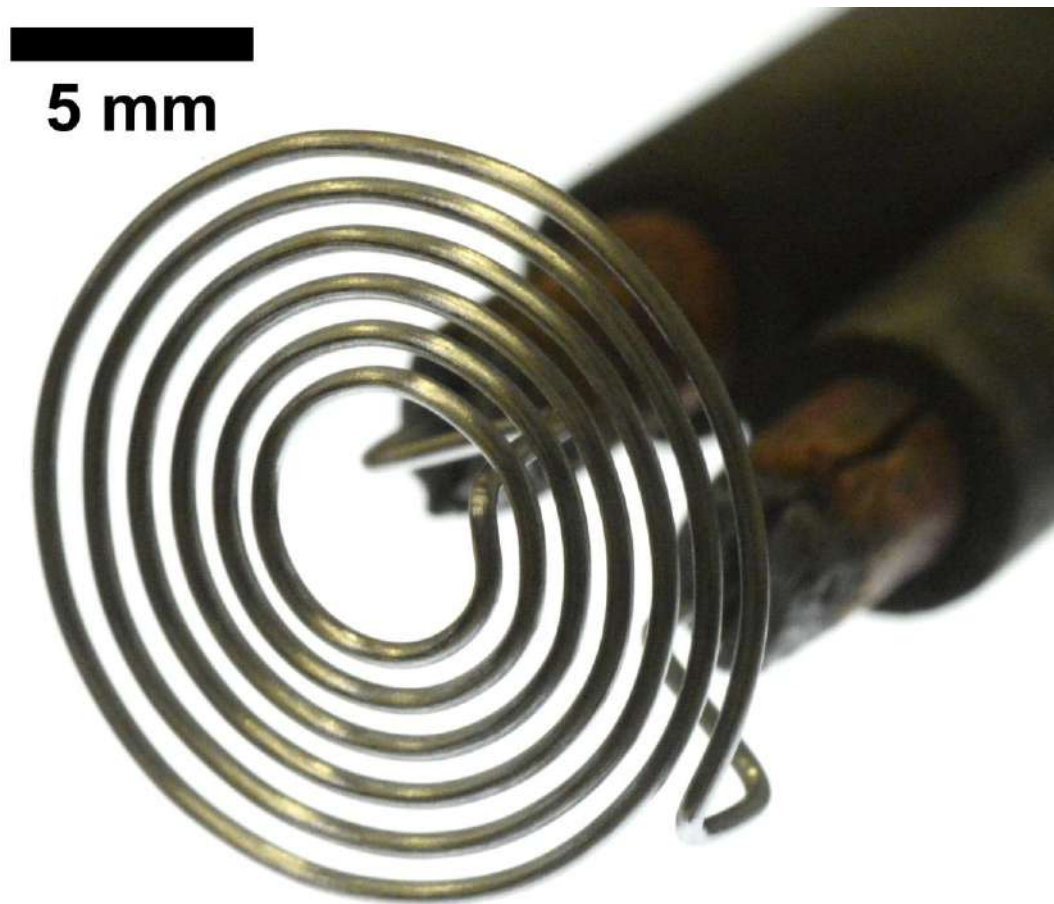
- Complex plasma-cathode interactions

Application



- Cooling hot surfaces by electron emission
(*Electron Transpiration Cooling*)

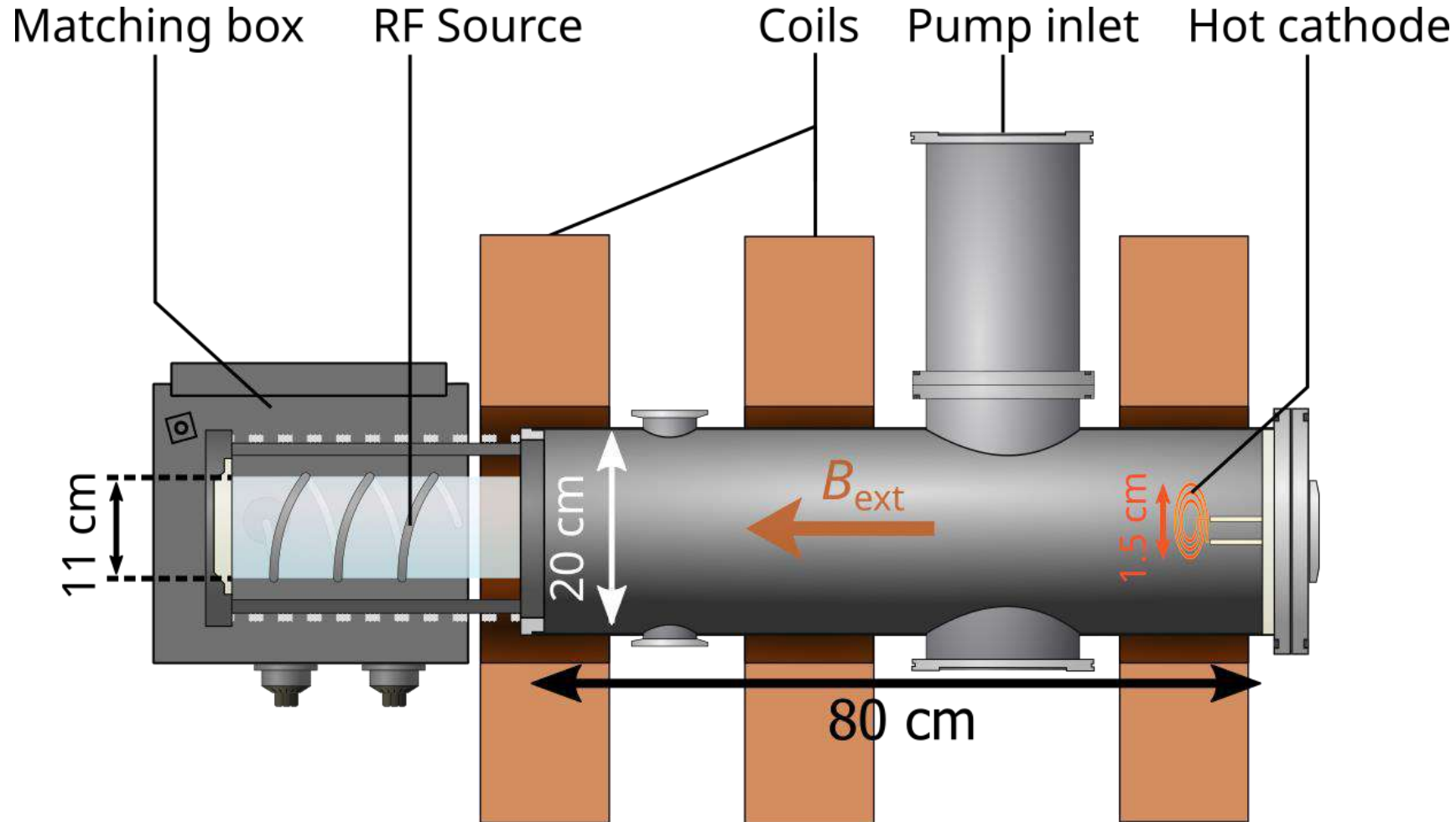
Pure tungsten cathode



- W emits 0.001 A/cm^2 at 2000 K , 10 A/cm^2 at 3000 K (Richardson current)
 - Very simple and low cost

Experimental setup: VKP

8



Argon plasma

$$p = 1 - 3 \text{ mTorr}$$

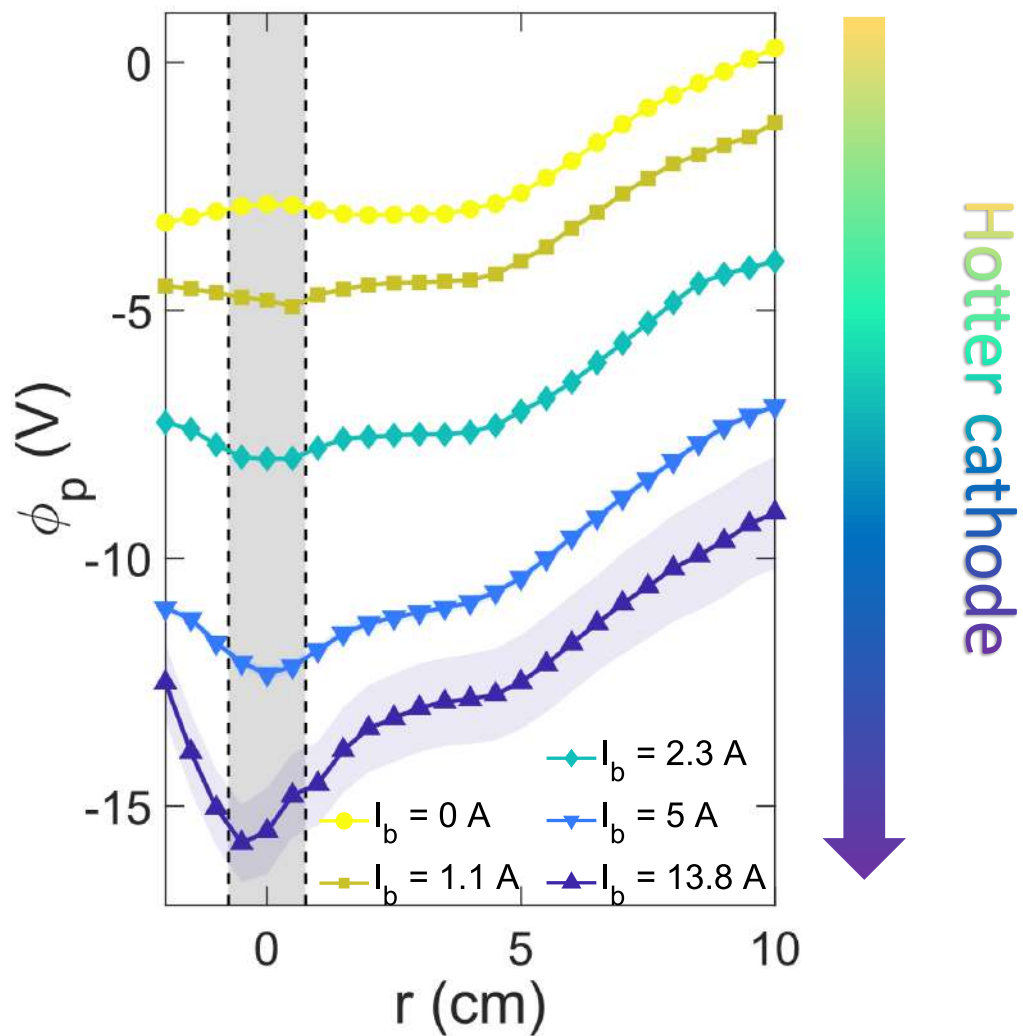
$$P_w = 1 \text{ kW}$$

$$B_{ext} = 170 \text{ G}$$

B_{ext} (G)	n (10^{18} m^{-3})	T_e (eV)	T_i (eV)	γ (%)	ρ_i (cm)	ρ_e (cm)
170	1.0	4.5	0.2	3.1	2.2	0.04
510	4.0	4.5	0.5	15.6	1.1	0.01

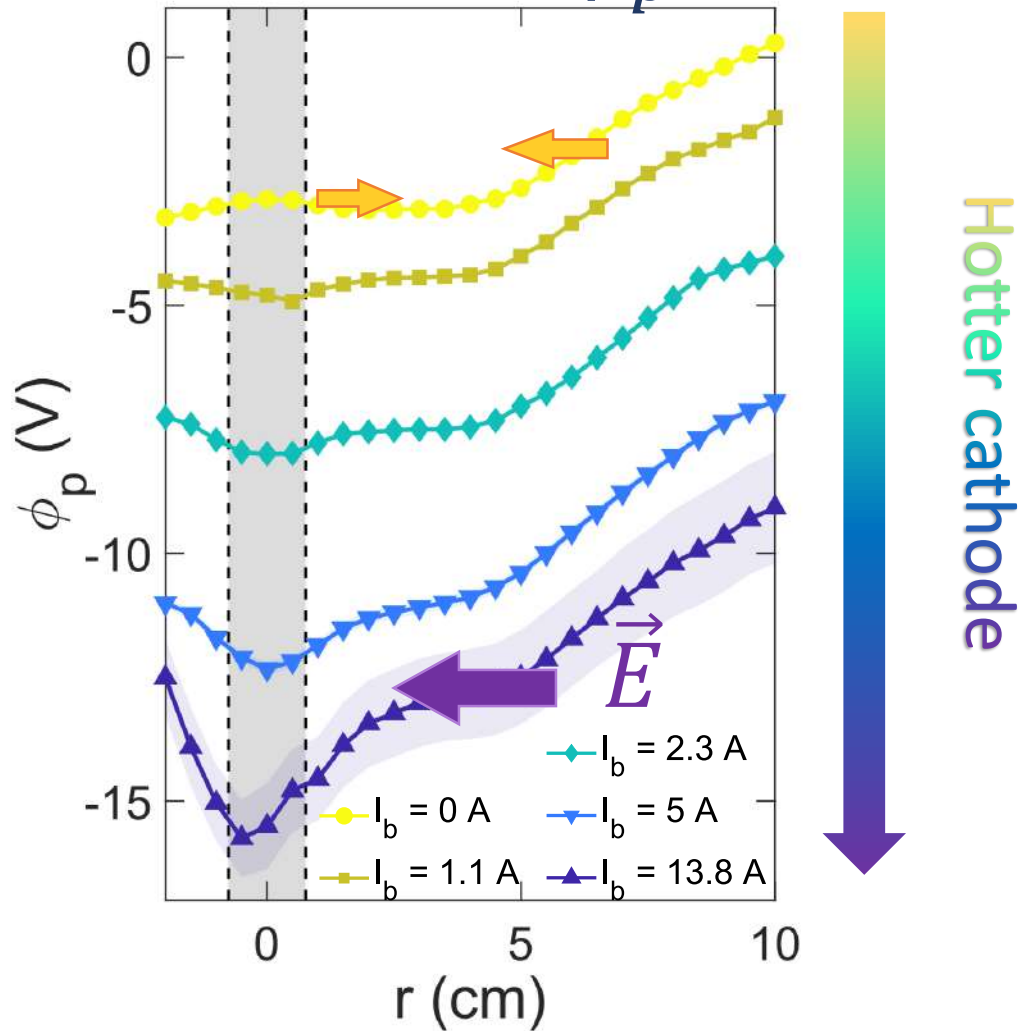


Plasma potential control



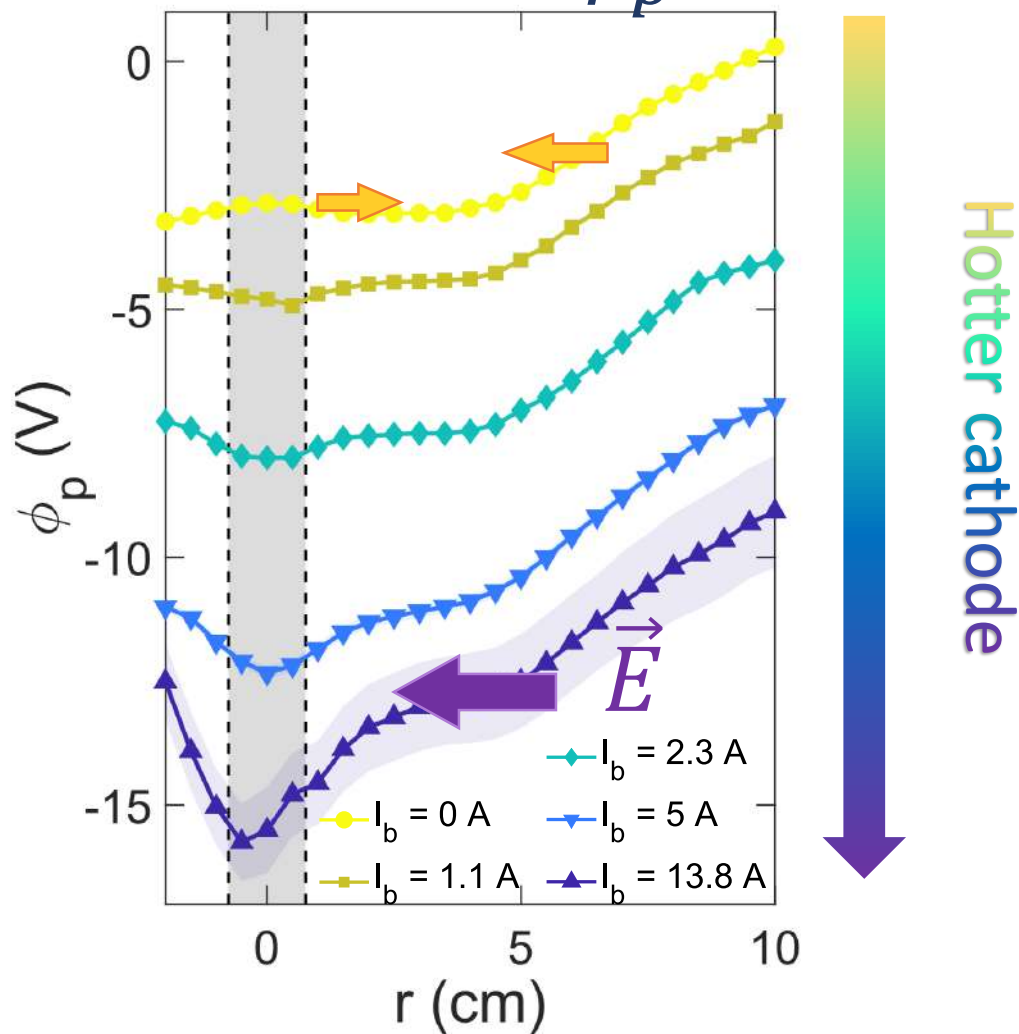
Plasma potential control

$$\vec{E} = -\vec{\nabla}\phi_p$$



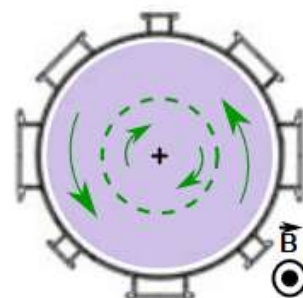
Plasma potential control

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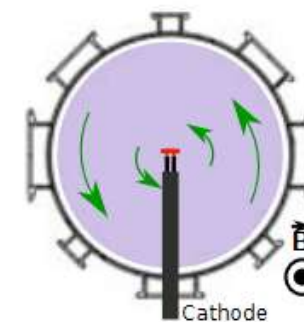


Goal: Control the plasma rotation

No Cathode



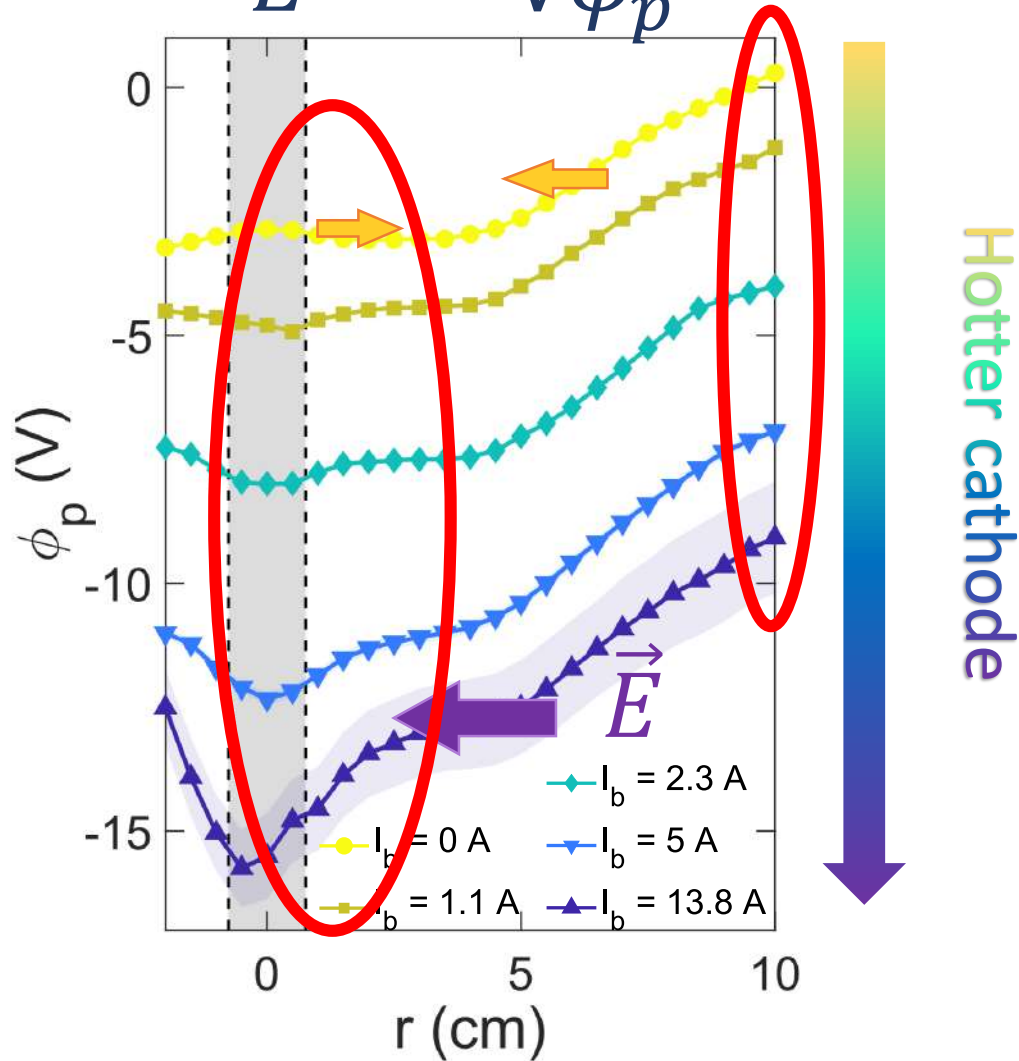
Cathode in the center



Plasma potential control

$$\vec{E} = -\vec{\nabla}\phi_p$$

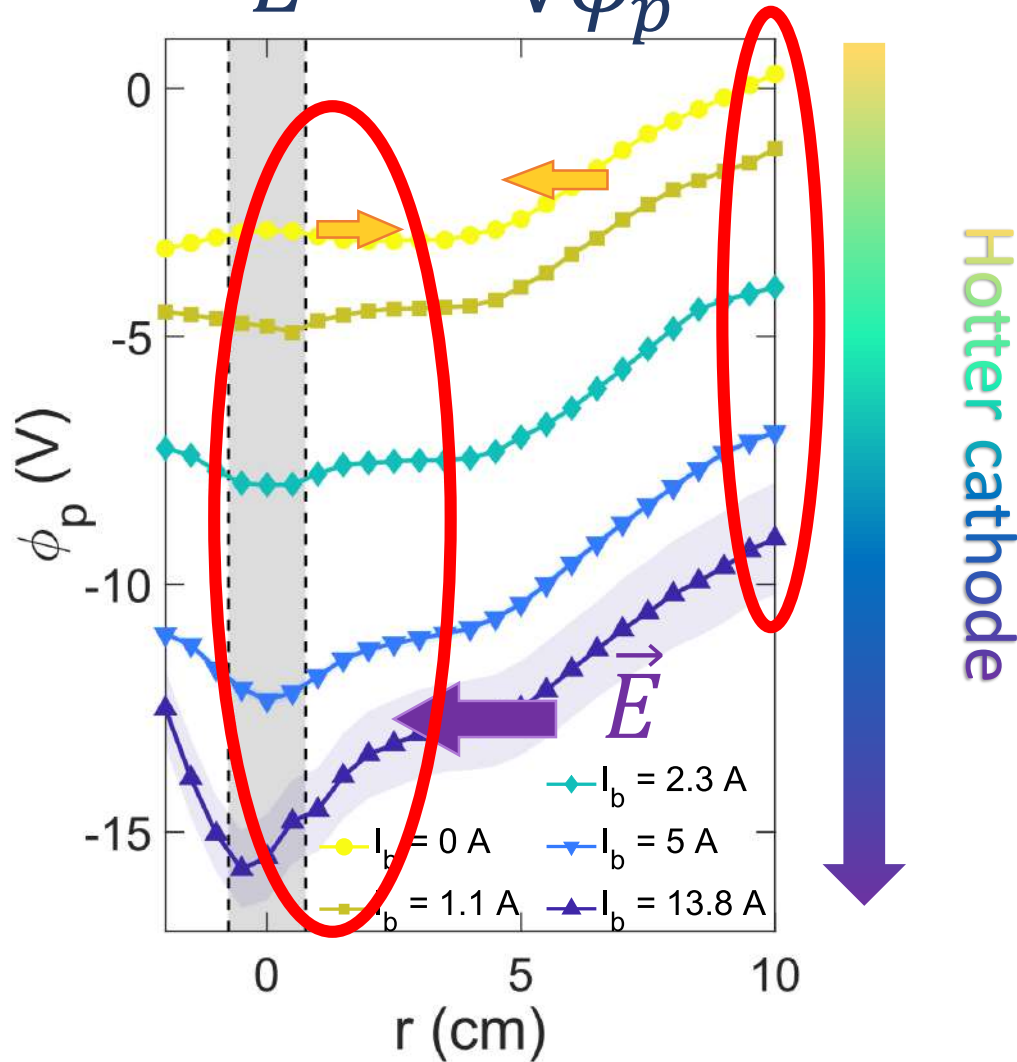
Poster: Nicolas Plihon, LTDP Thursday session



Plasma potential control

Poster: Nicolas Plihon, LTDP Thursday session

$$\vec{E} = -\vec{\nabla}\phi_p$$



Modelling requires
accurate I_b prediction

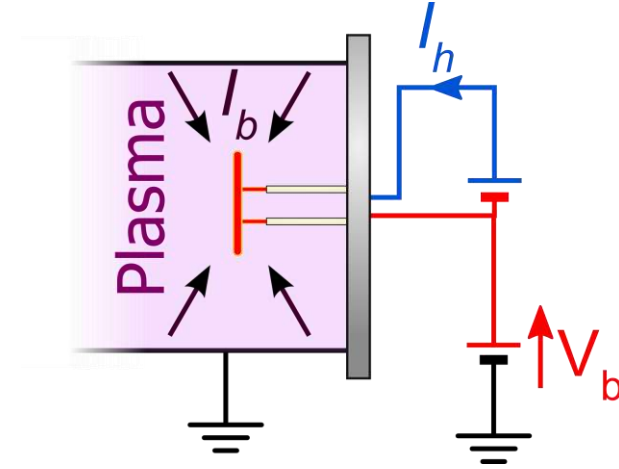
OUTLINE

1. Introduction
2. Spatially and temporally resolved T_W measurements
3. Detailed thermal model
4. Applications

Role of the cathode bias

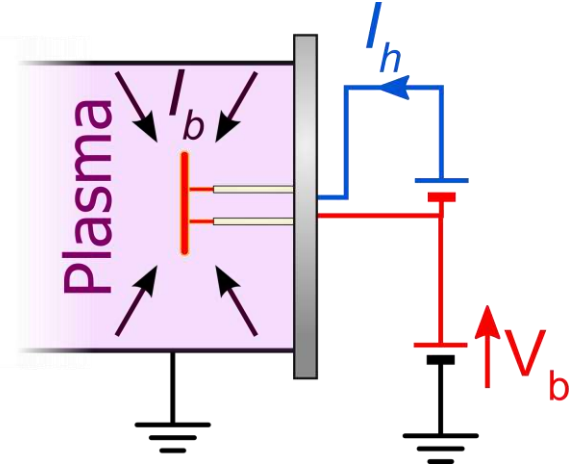
$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e}$$

$I_i, I_e \ll I_{em}$



Role of the cathode bias

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e} \quad \xrightarrow{I_i, I_e \ll I_{em}} \quad I_{em} = A_G S T_W^2 \exp \left(-\frac{eW}{k_B T_W} \right)$$

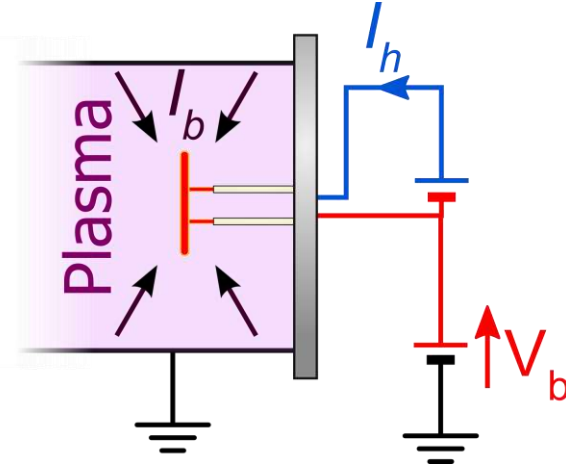


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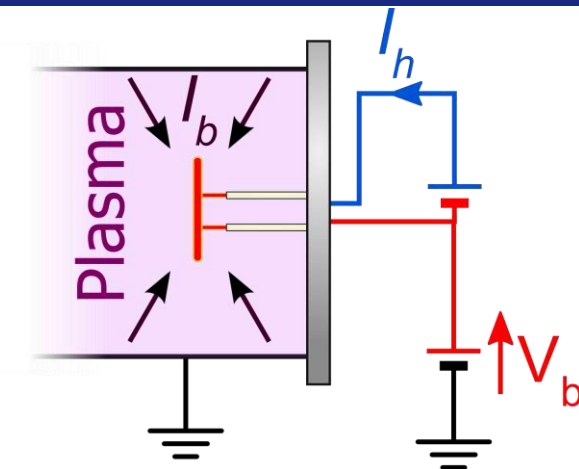
Naive prediction

- I_h controls I_b
- No role of V_b



Role of the cathode bias

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e} \quad \xrightarrow{I_i, I_e \ll I_{em}} \quad I_{em} = A_G S T_W^2 \exp\left(-\frac{eW}{k_B T_W}\right)$$

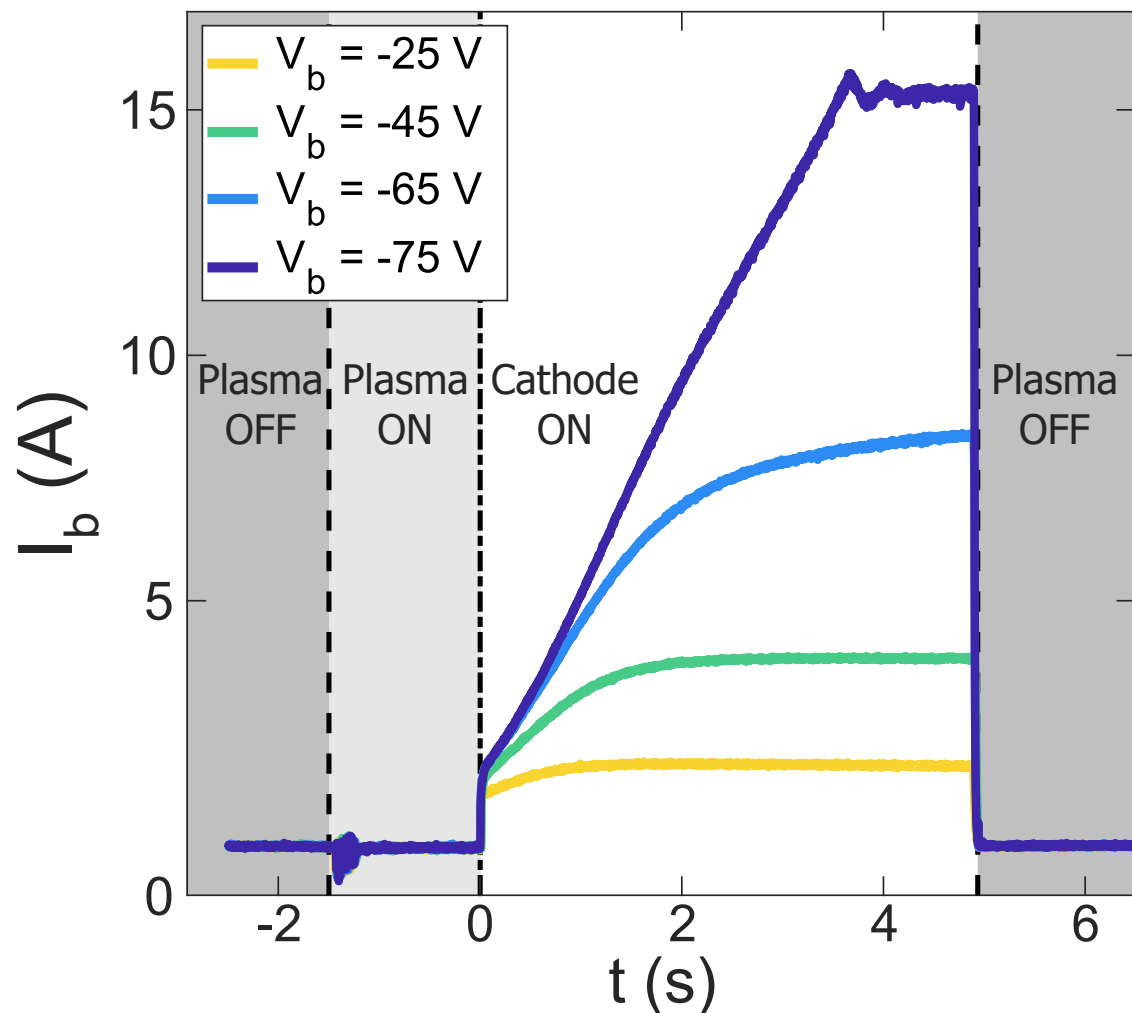


Naive prediction

- I_h controls I_b
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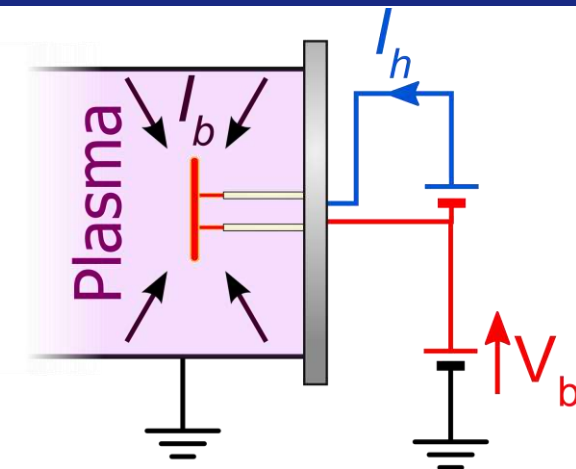
At fixed I_h and varying V_b :

- I_b depends on V_b
- Runaway behaviour at $V_b \ll \phi_p$



Role of the cathode bias

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e} \quad \xrightarrow{I_i, I_e \ll I_{em}} \quad I_{em} = A_G S T_W^2 \exp\left(-\frac{eW}{k_B T_W}\right)$$



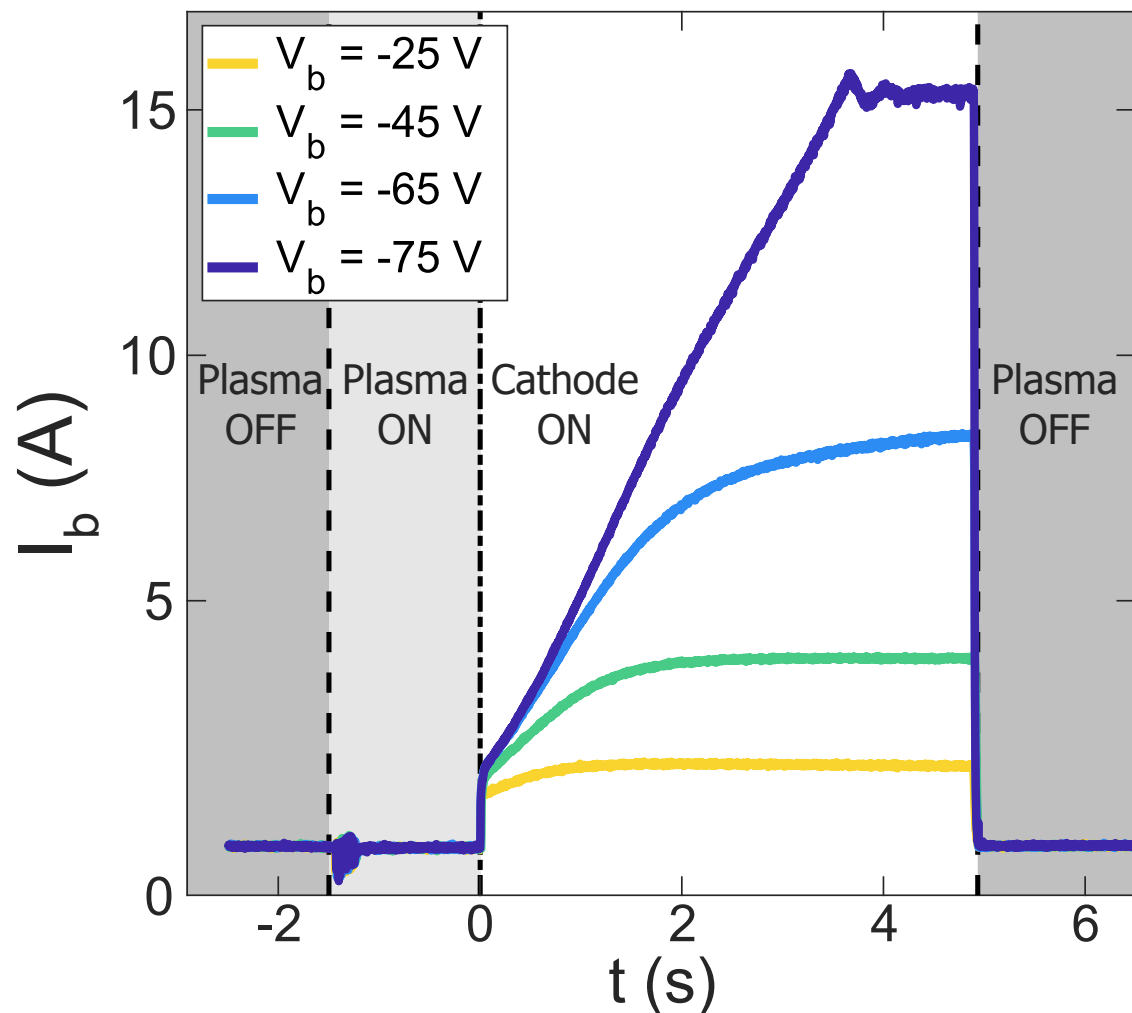
Naive prediction

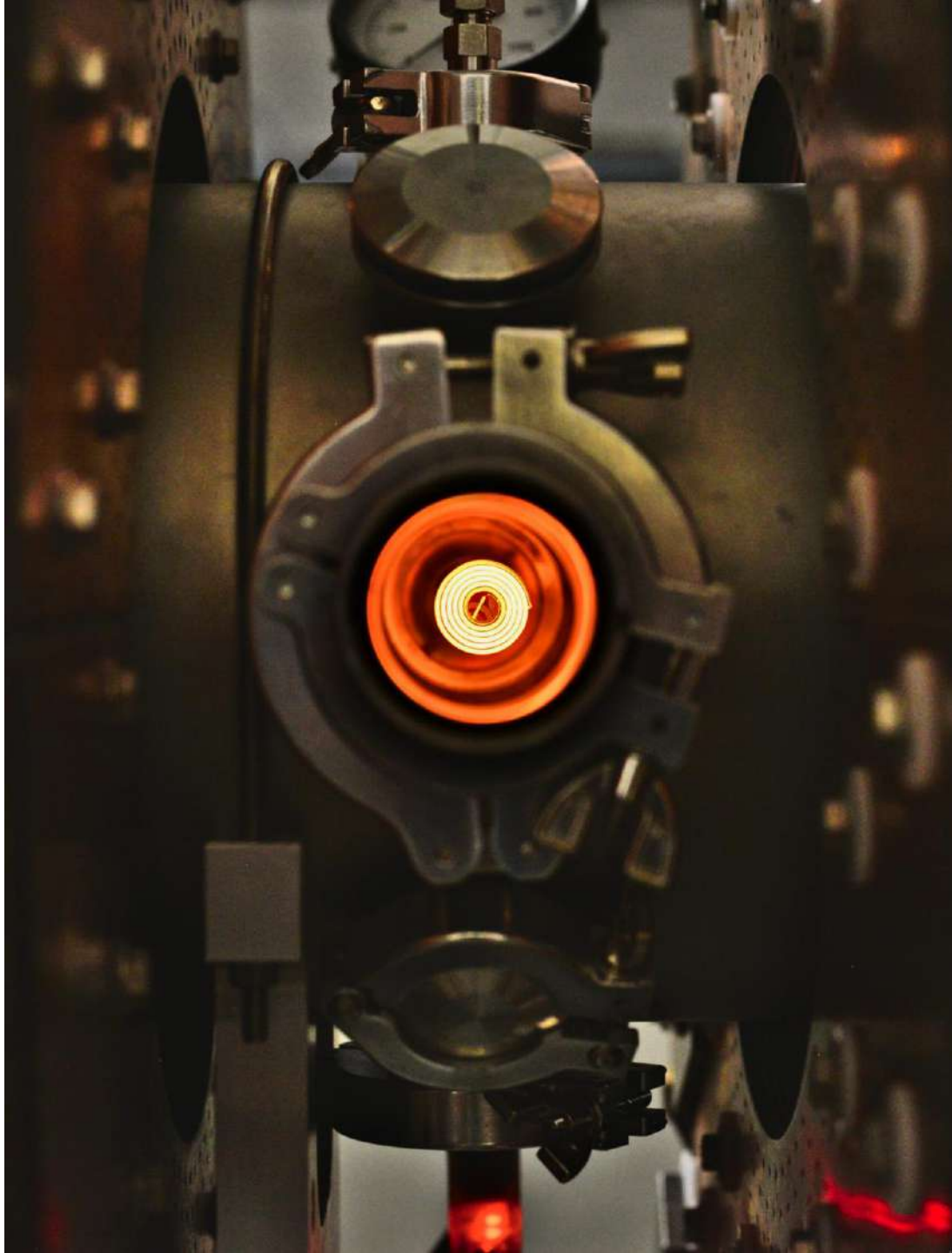
- I_h controls I_b
- No role of V_b

At fixed I_h and varying V_b :

- I_b depends on V_b
- Runaway behaviour at $V_b \ll \phi_p$

How T_W depends on I_h, V_b ...

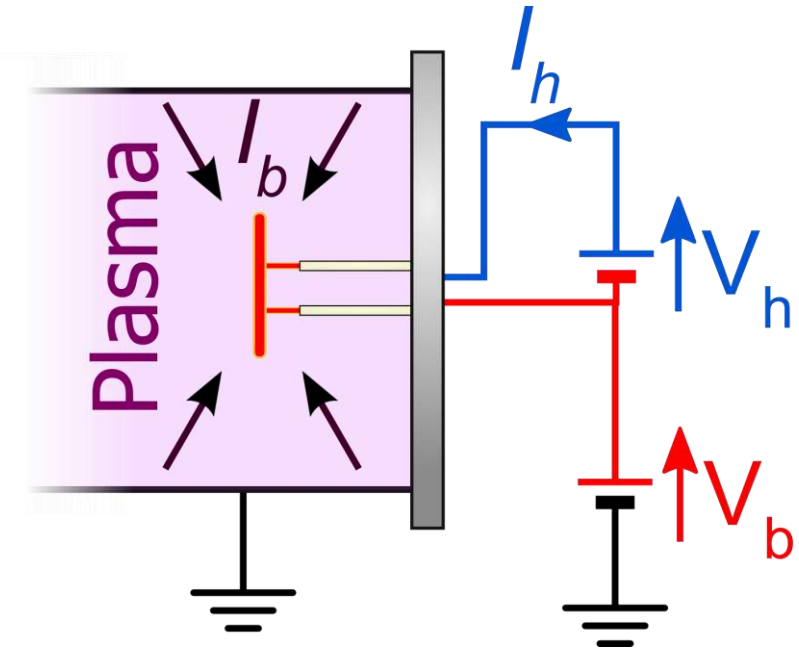
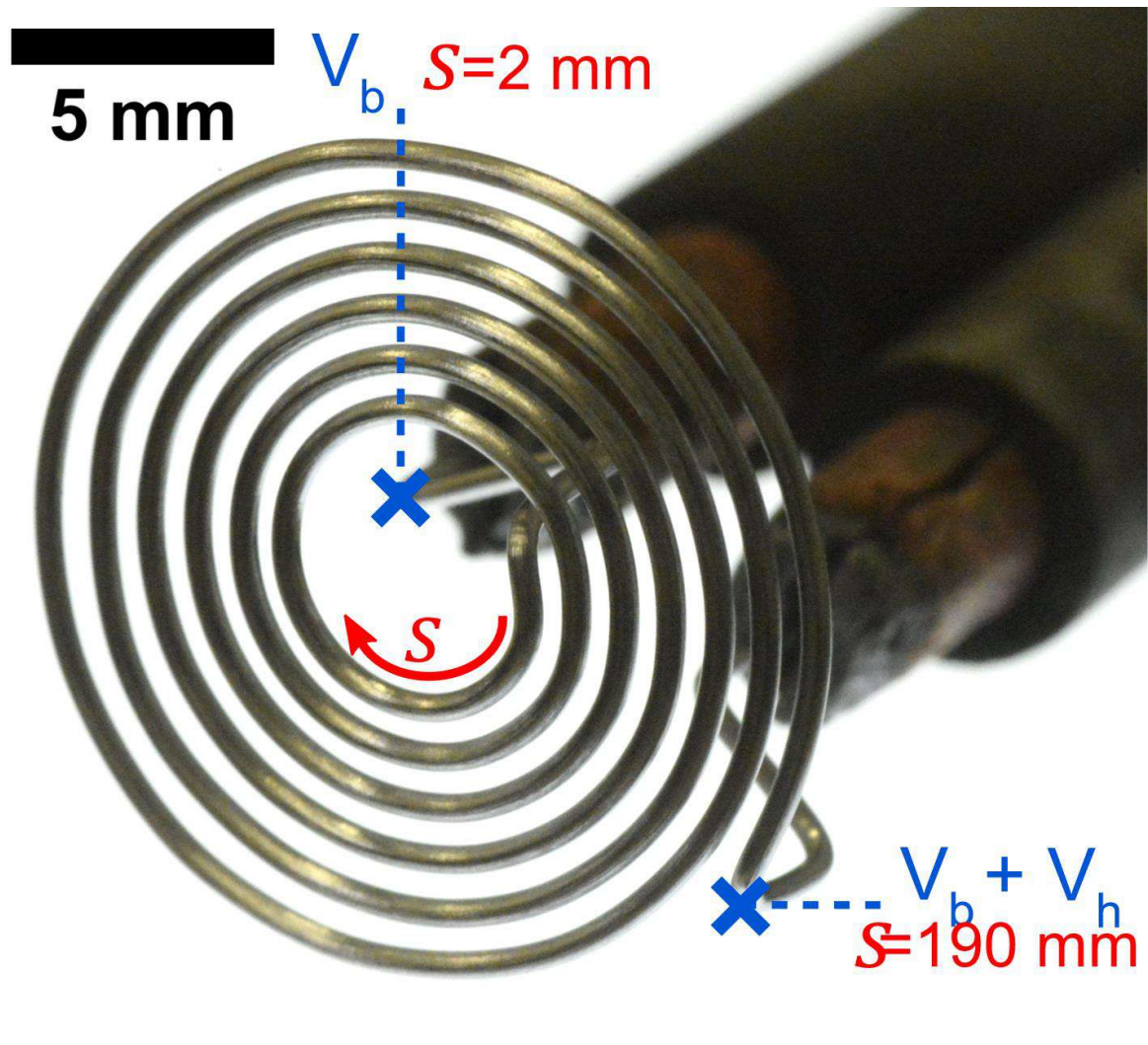




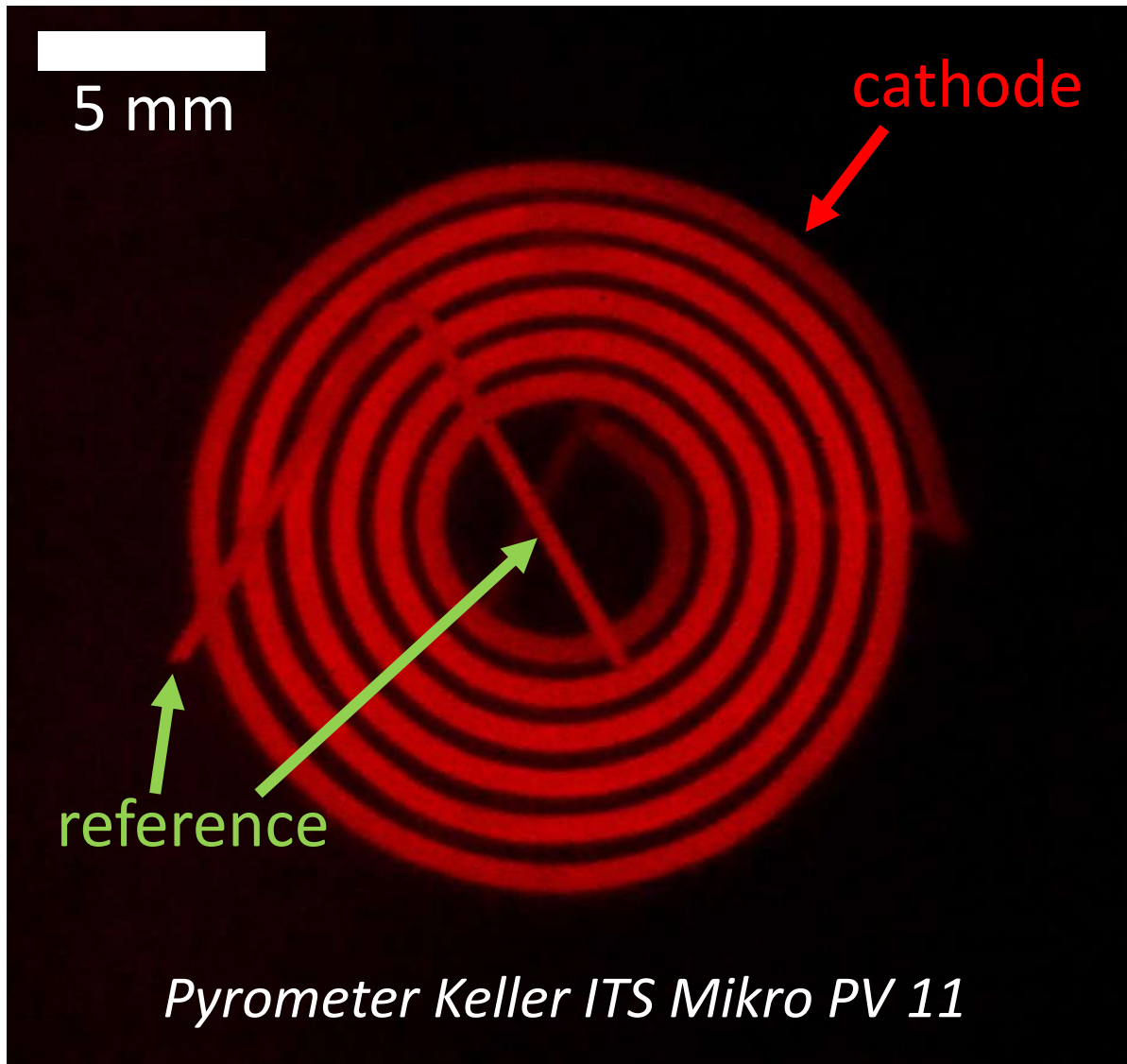
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Spatially and temporally resolved T_W measurements

Formal description of the cathode

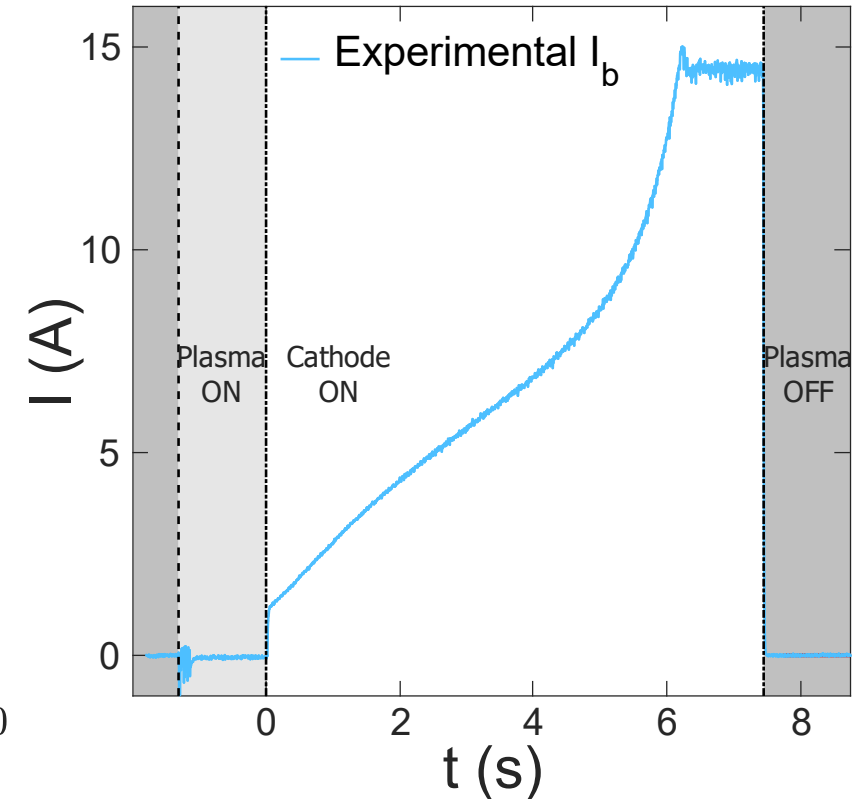
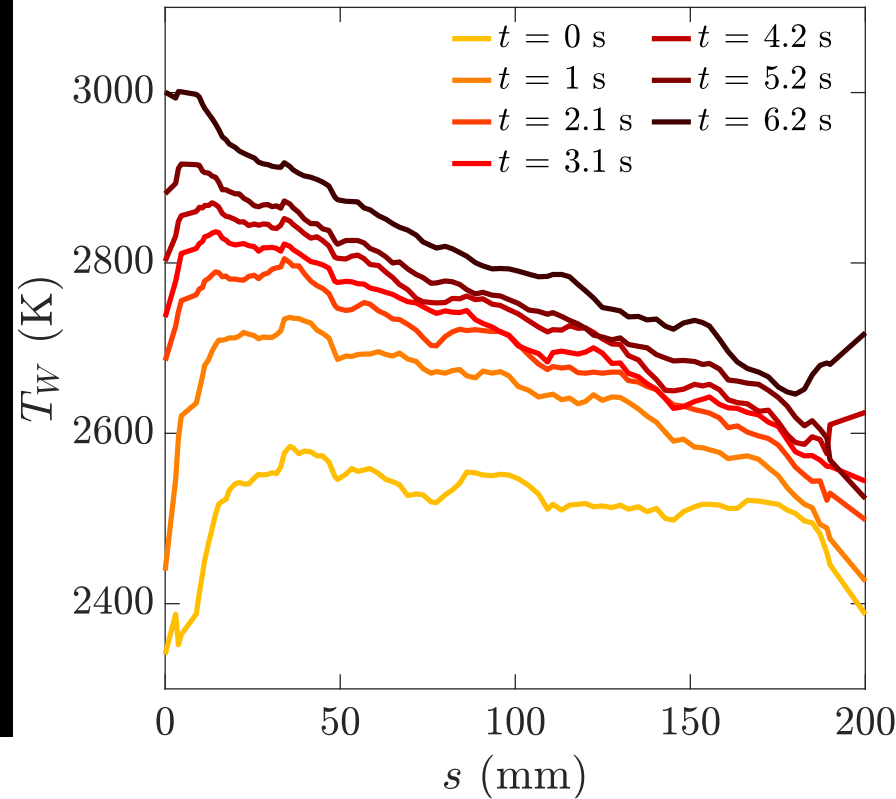


- Curvilinear abscissa of the cathode s
- Potential from V_b to $V_b + V_h < \phi_p$



- Light intensity comparison at 650 nm between cathode and a reference gray body.
- **Measuring T_W**

Heterogeneous temperature profile



- Plasma transparent at 650 nm
 - **Heterogeneous heating**
 - Runaway behaviour

Computing thermionic emission

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e}$$

$I_i, I_e \ll I_{em}$

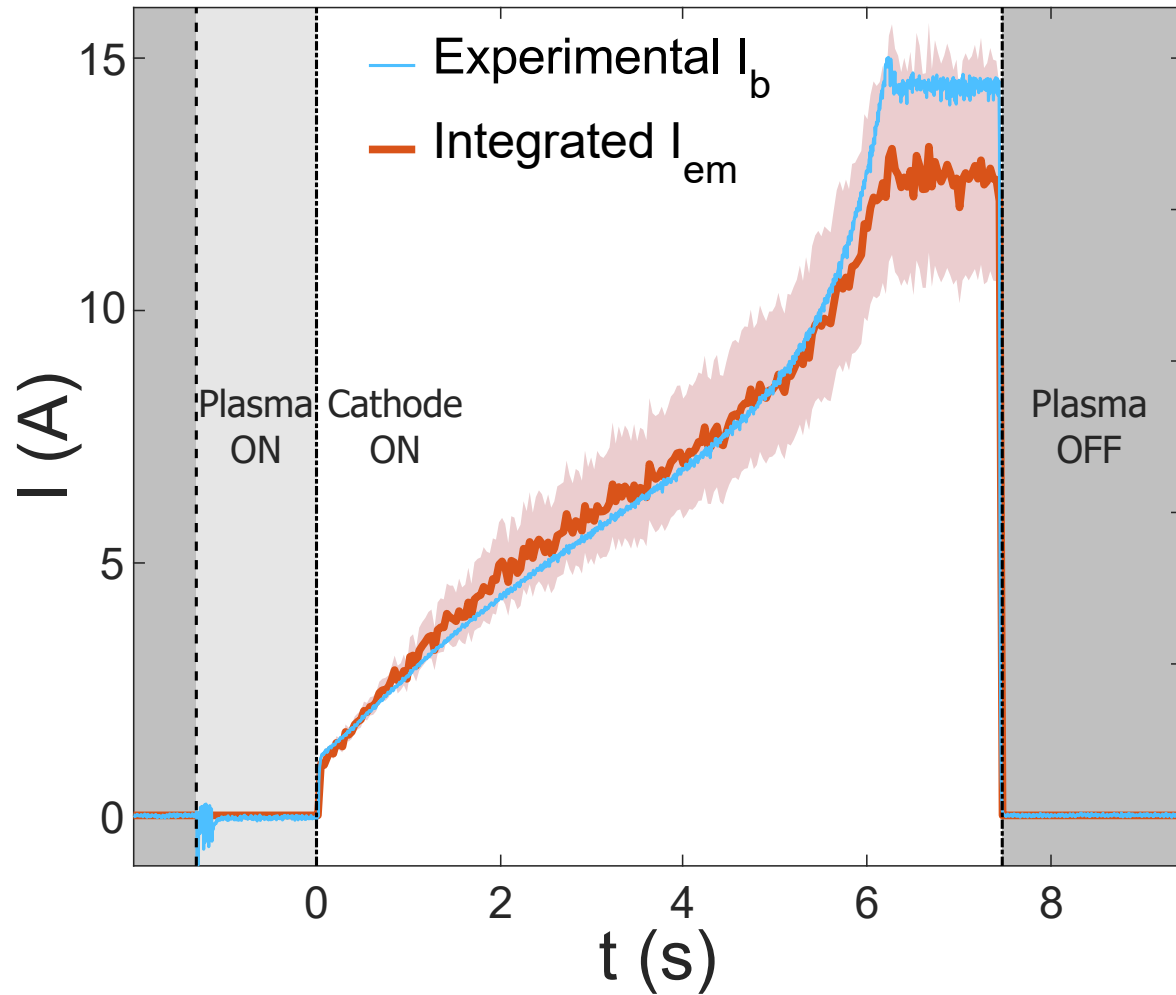
$$I_{em} = \int_0^s A_G T_W^2 \exp\left(-\frac{eW}{k_B T_W}\right) 2\pi r \, dx$$

Computing thermionic emission

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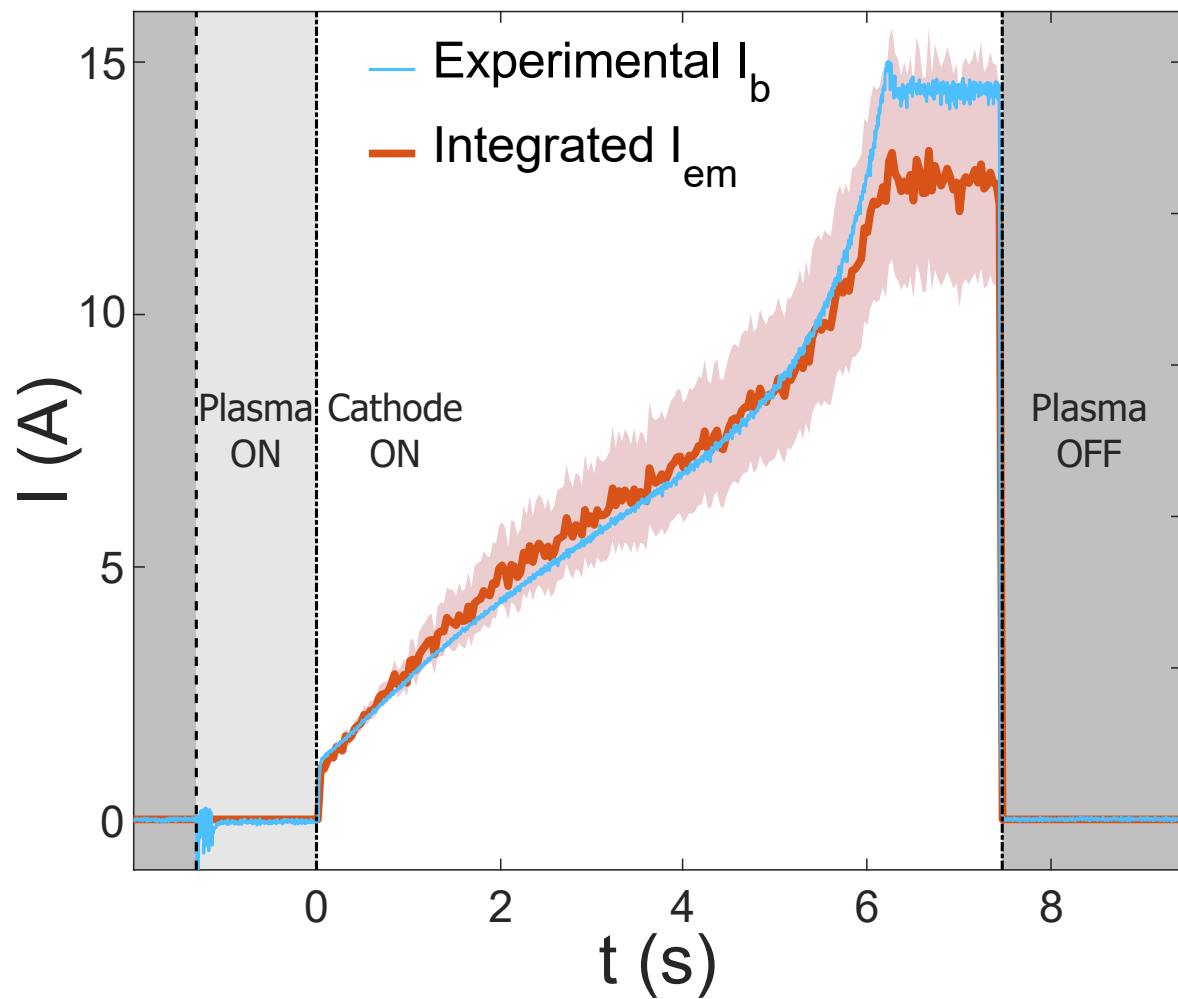


Computing thermionic emission

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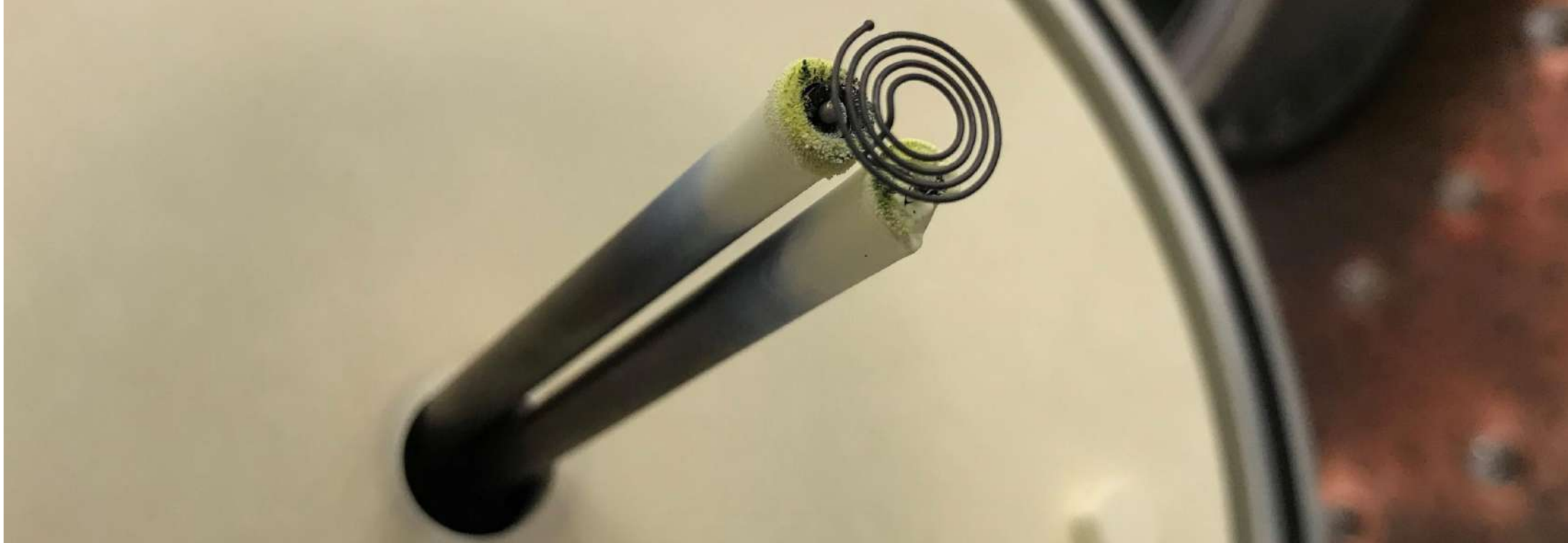
$I_i, I_e \ll I_{em}$

$$I_{em} = \int_0^s A_G T_W^2 \exp\left(-\frac{eW}{k_B T_W}\right) 2\pi r dx$$



➤ I_{em} well predicted by T_W measurements.

➔ *Modelling T_W profile accurately?*



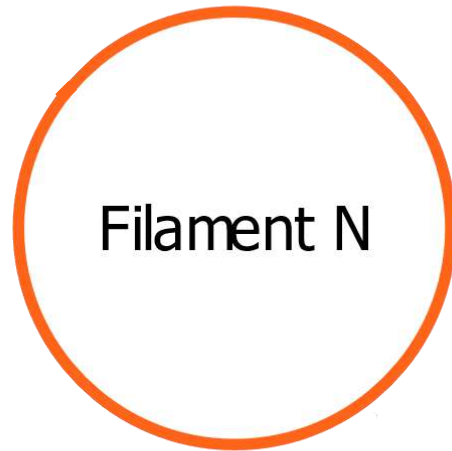
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Detailed thermal model

Heat budget equation

$$\dot{H} =$$

$$\dot{H} = c_p \rho V \frac{\partial T_w}{\partial t}$$

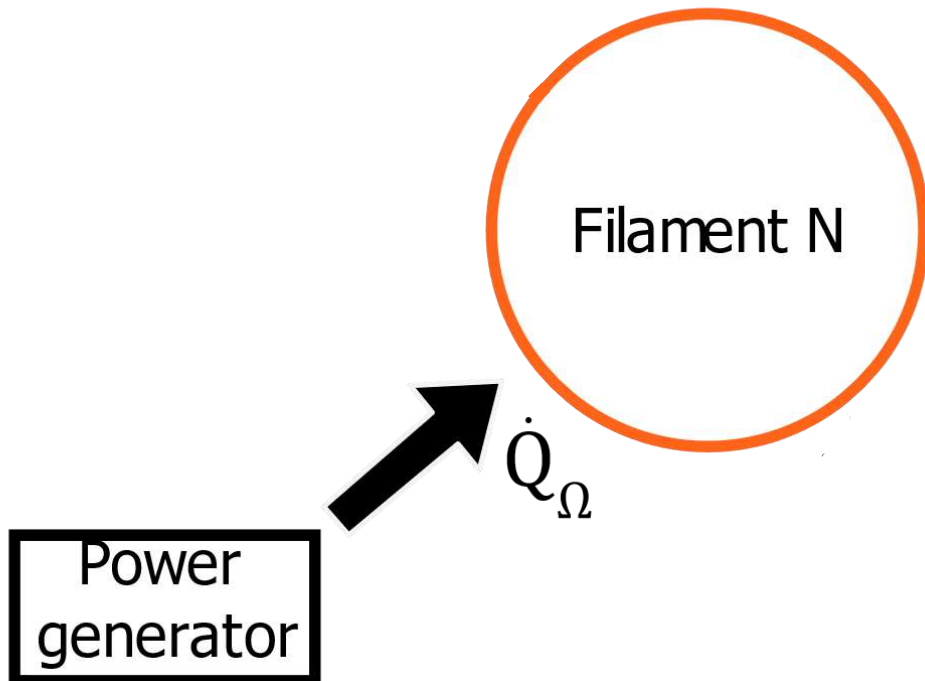


Heat budget equation

$$\dot{H} = \dot{Q}_{\Omega}$$

$$\dot{H} = c_p \rho V \frac{\partial T_w}{\partial t}$$

$$\dot{Q}_{\Omega} = R (I_h + I_b)^2$$



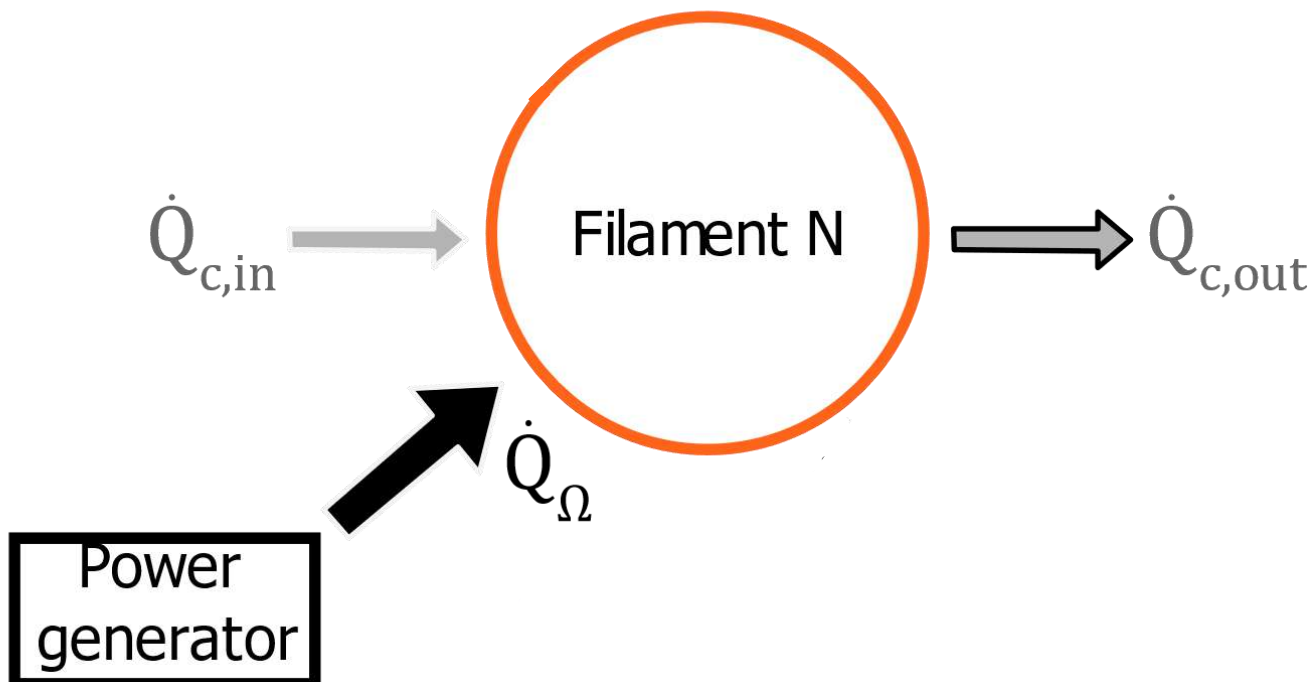
Heat budget equation

$$\dot{H} = \dot{Q}_\Omega + \dot{Q}_c$$

$$\dot{H} = C_p \rho V \frac{\partial T_w}{\partial t}$$

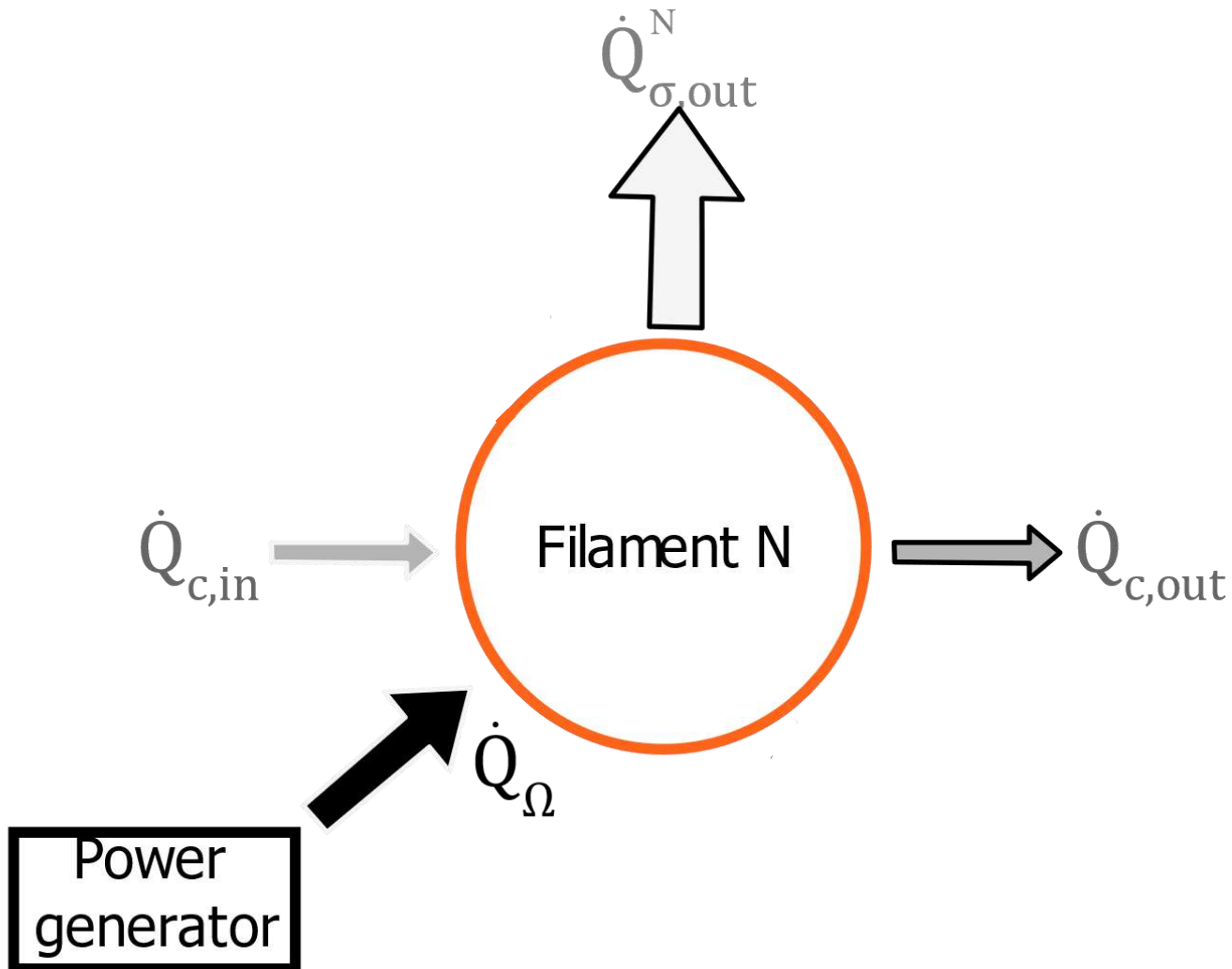
$$\dot{Q}_\Omega = R (I_h + I_b)^2$$

$$\dot{Q}_c = \lambda \pi r^2 \frac{\partial^2 T_w}{\partial s^2}$$



Heat budget equation

$$\dot{H} = \dot{Q}_{\Omega} + \dot{Q}_c - \dot{Q}_{\sigma,out}$$



$$\dot{H} = C_p \rho V \frac{\partial T_W}{\partial t}$$

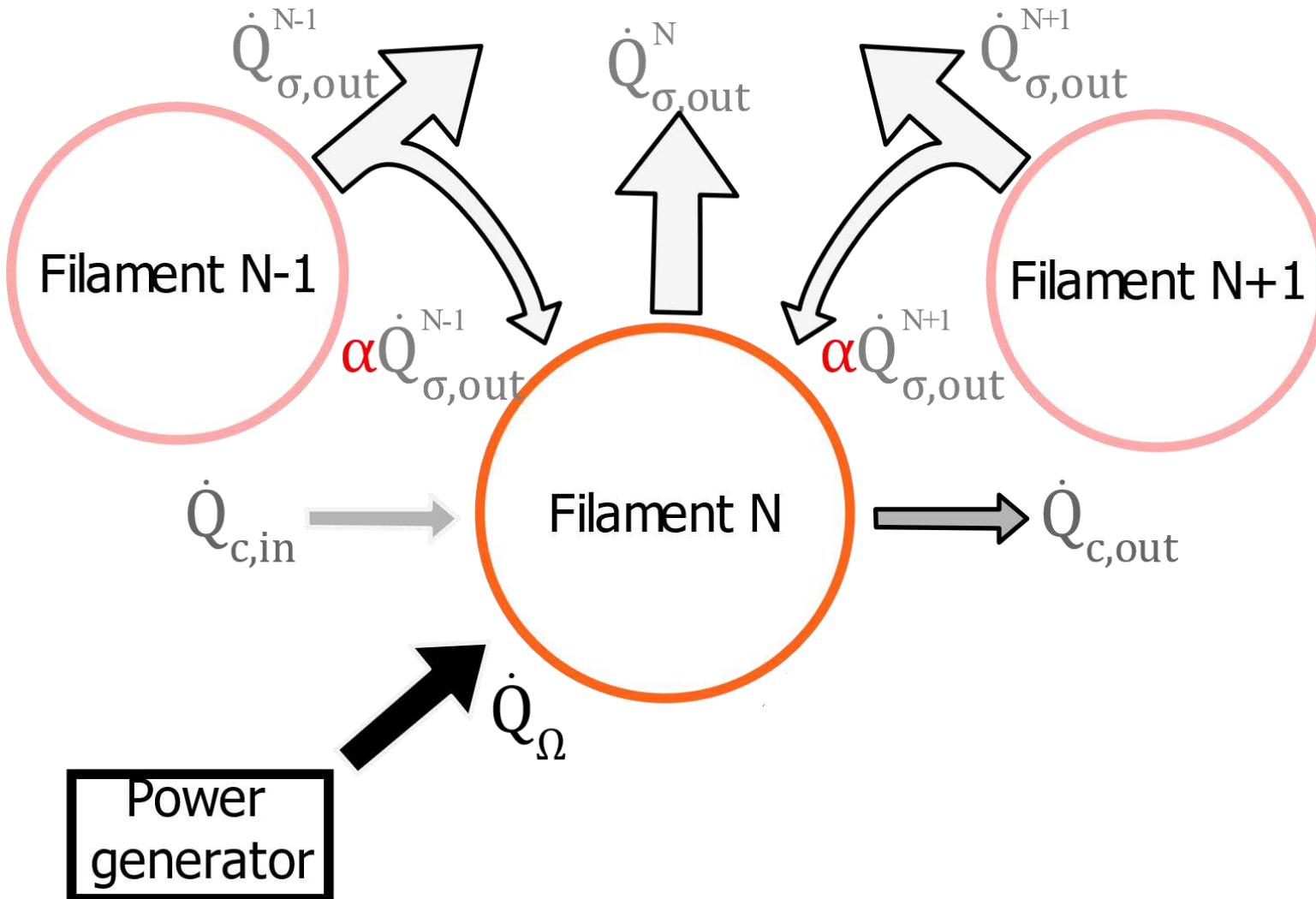
$$\dot{Q}_{\Omega} = R (I_h + I_b)^2$$

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$$\dot{Q}_{\sigma,out} = \sigma S \epsilon T_W^4$$

Heat budget equation

$$\dot{H} = \dot{Q}_{\Omega} + \dot{Q}_c - \dot{Q}_{\sigma,out} + \dot{Q}_{\sigma,in}$$



$$\dot{H} = C_p \rho V \frac{\partial T_w}{\partial t}$$

$$\dot{Q}_{\Omega} = R (I_h + I_b)^2$$

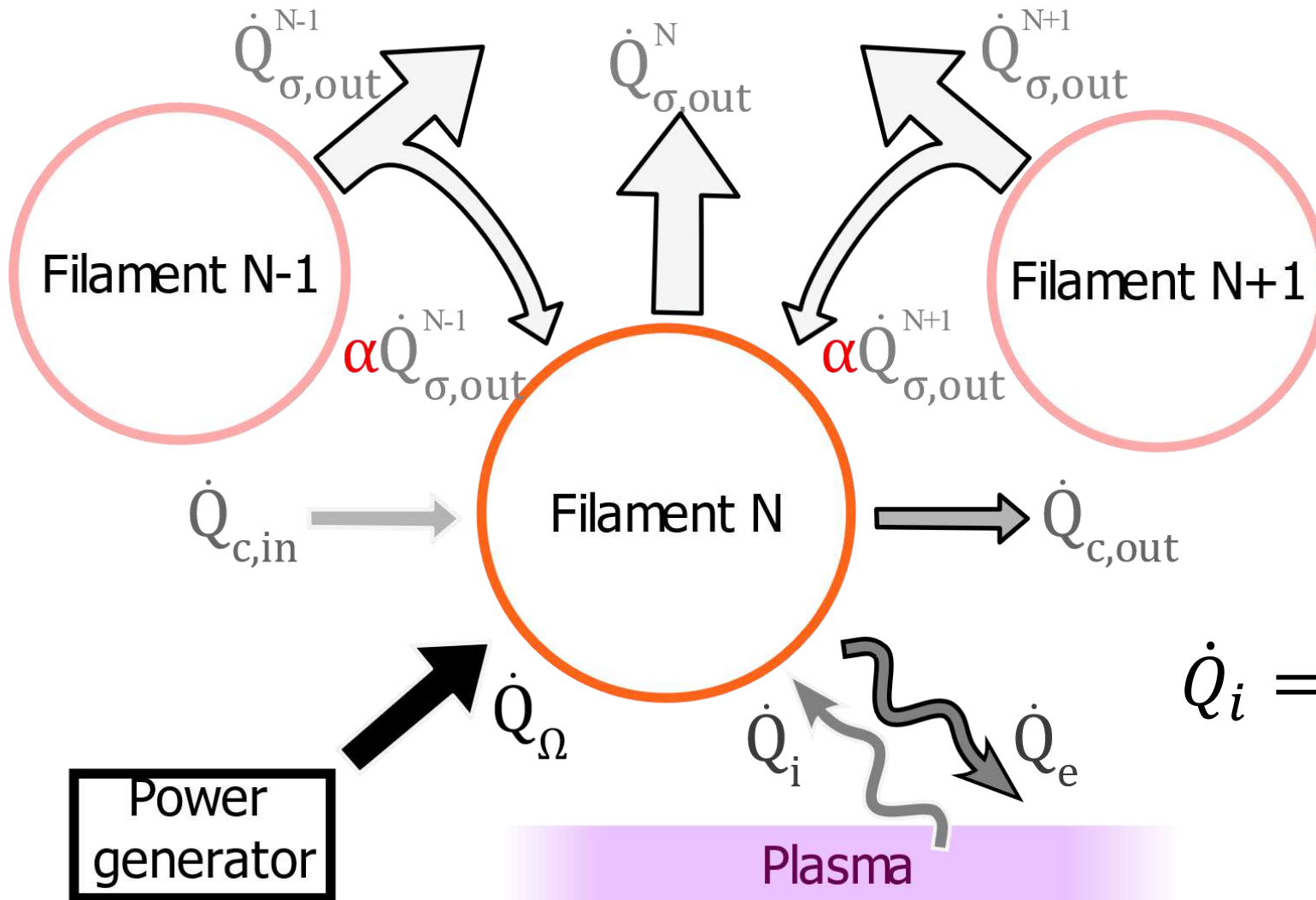
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$$\dot{Q}_{\sigma,out} = \sigma S \epsilon T_w^4$$

$$\dot{Q}_{\sigma,in} = \alpha (Q_{\sigma,out}^{N-1} + Q_{\sigma,out}^{N+1})$$

Heat budget equation

$$\dot{H} = \dot{Q}_\Omega + \dot{Q}_c - \dot{Q}_{\sigma,out} + \dot{Q}_{\sigma,in} + \dot{Q}_i - \dot{Q}_e$$



$$\dot{H} = C_p \rho V \frac{\partial T_W}{\partial t}$$

$$\dot{Q}_\Omega = R (I_h + I_b)^2$$

$$\dot{Q}_c = \lambda \pi r^2 \frac{\partial^2 T_W}{\partial s^2}$$

$$\dot{Q}_{\sigma,out} = \sigma S \epsilon T_W^4$$

$$\dot{Q}_{\sigma,in} = \alpha (\dot{Q}_{\sigma,out}^{N-1} + \dot{Q}_{\sigma,out}^{N+1})$$

$$\dot{Q}_i = \frac{I_i}{e} e [E_i - W + \beta (\phi_p - V_b)]$$

$$\dot{Q}_e = \frac{I_{em}}{e} (eW + 2k_B T_W)$$

Simulation/experiments comparison

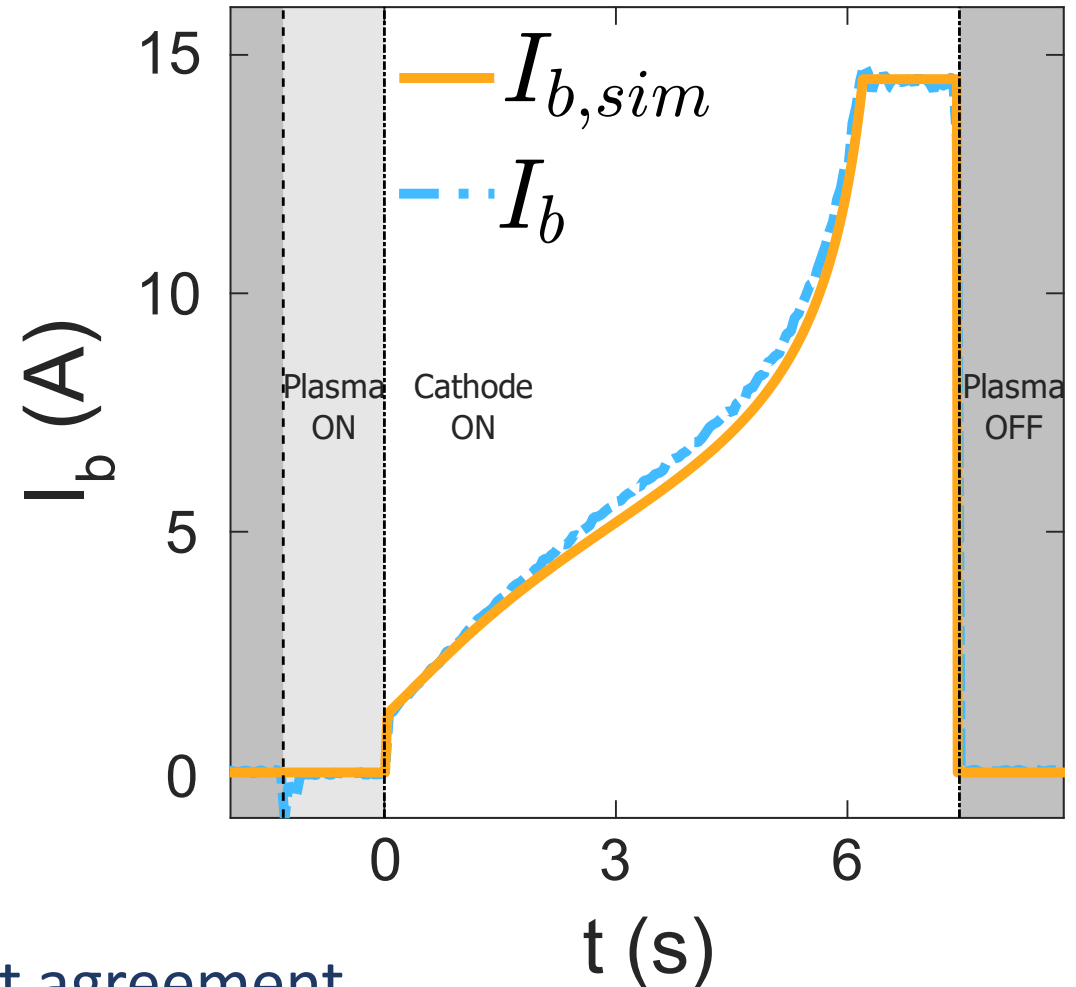
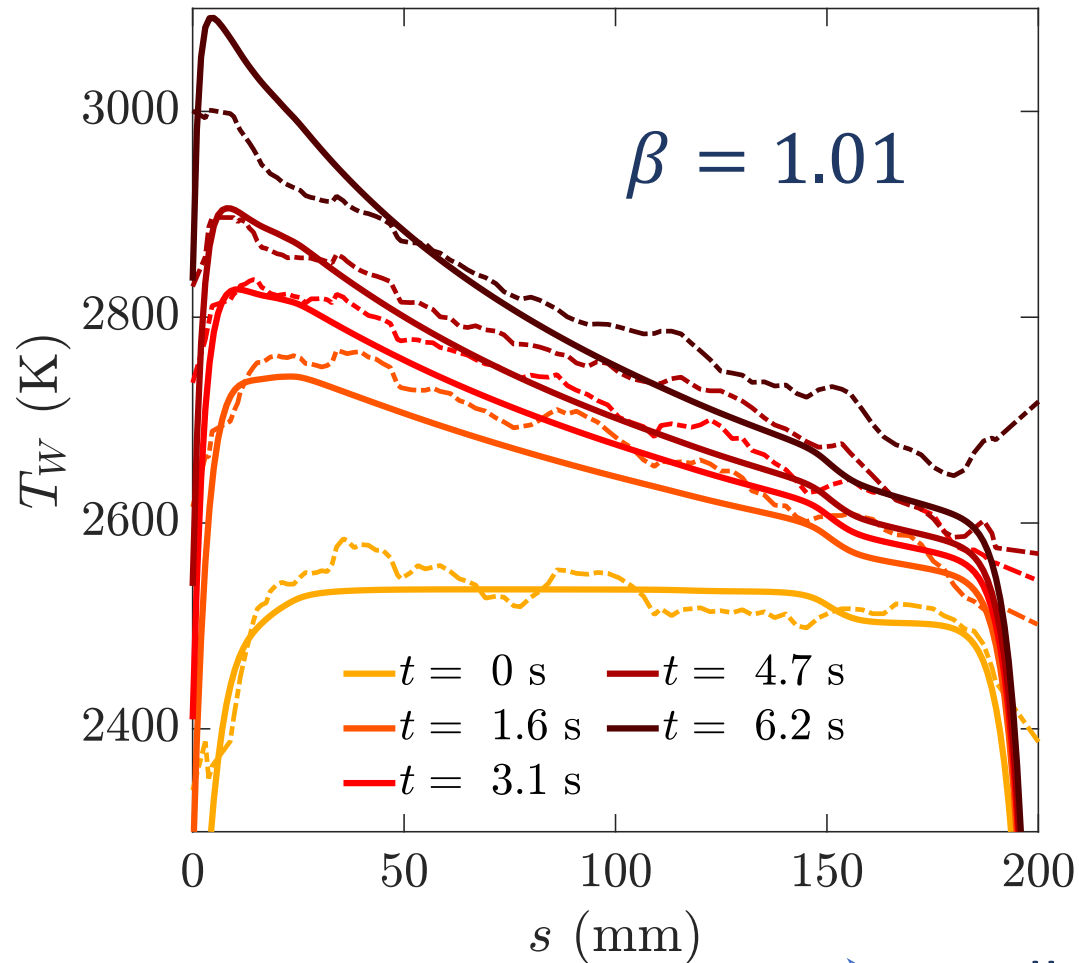
35

Numerical computation: $\dot{H} = \dot{Q}_{\Omega} + \dot{Q}_c - \dot{Q}_{\sigma,out} + \dot{Q}_{\sigma,in} + \dot{Q}_i - \dot{Q}_e$

Simulation/experiments comparison

36

Numerical computation: $\dot{H} = \dot{Q}_\Omega + \dot{Q}_c - \dot{Q}_{\sigma,out} + \dot{Q}_{\sigma,in} + \dot{Q}_i - \dot{Q}_e$



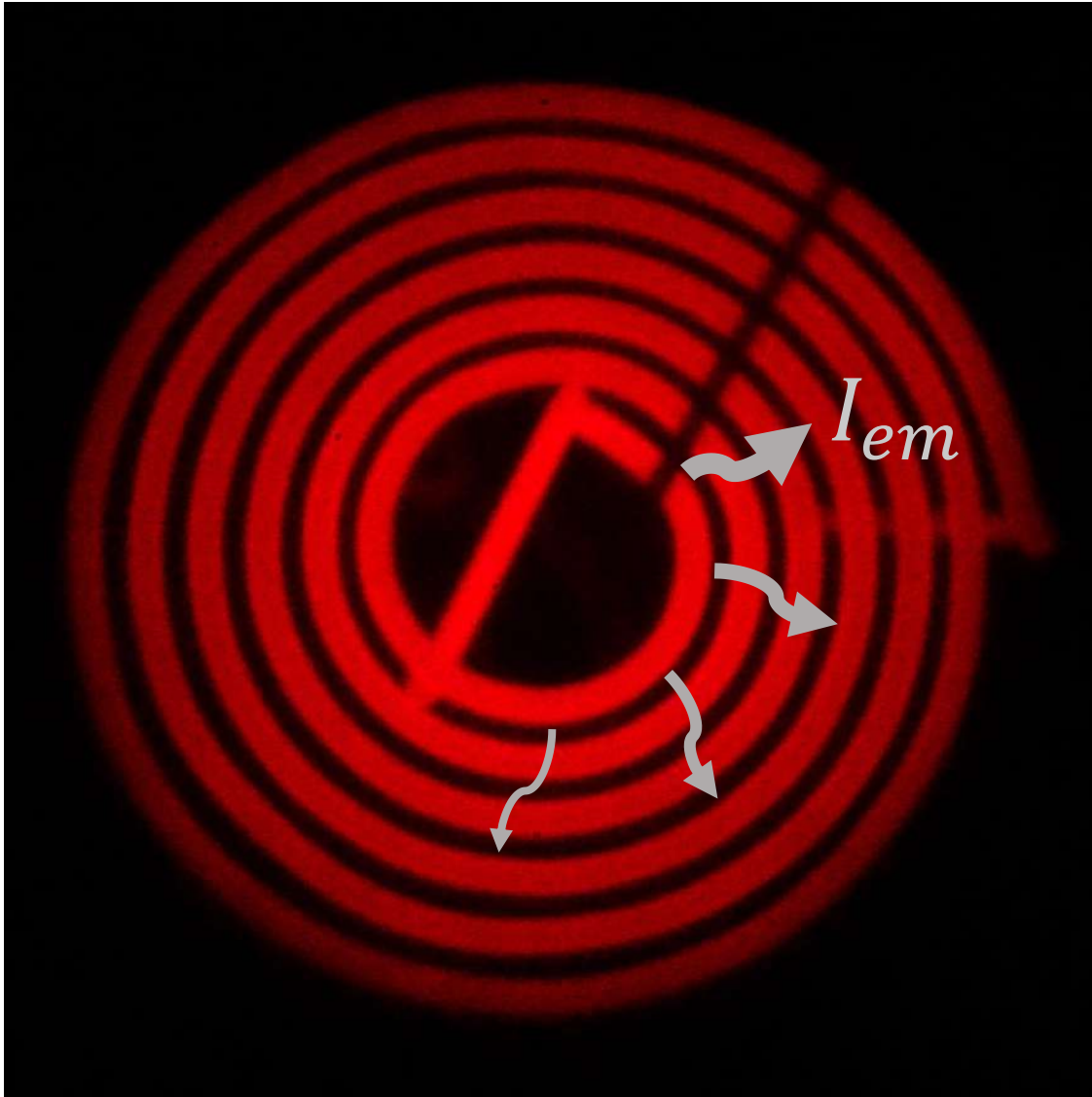
➤ Excellent agreement

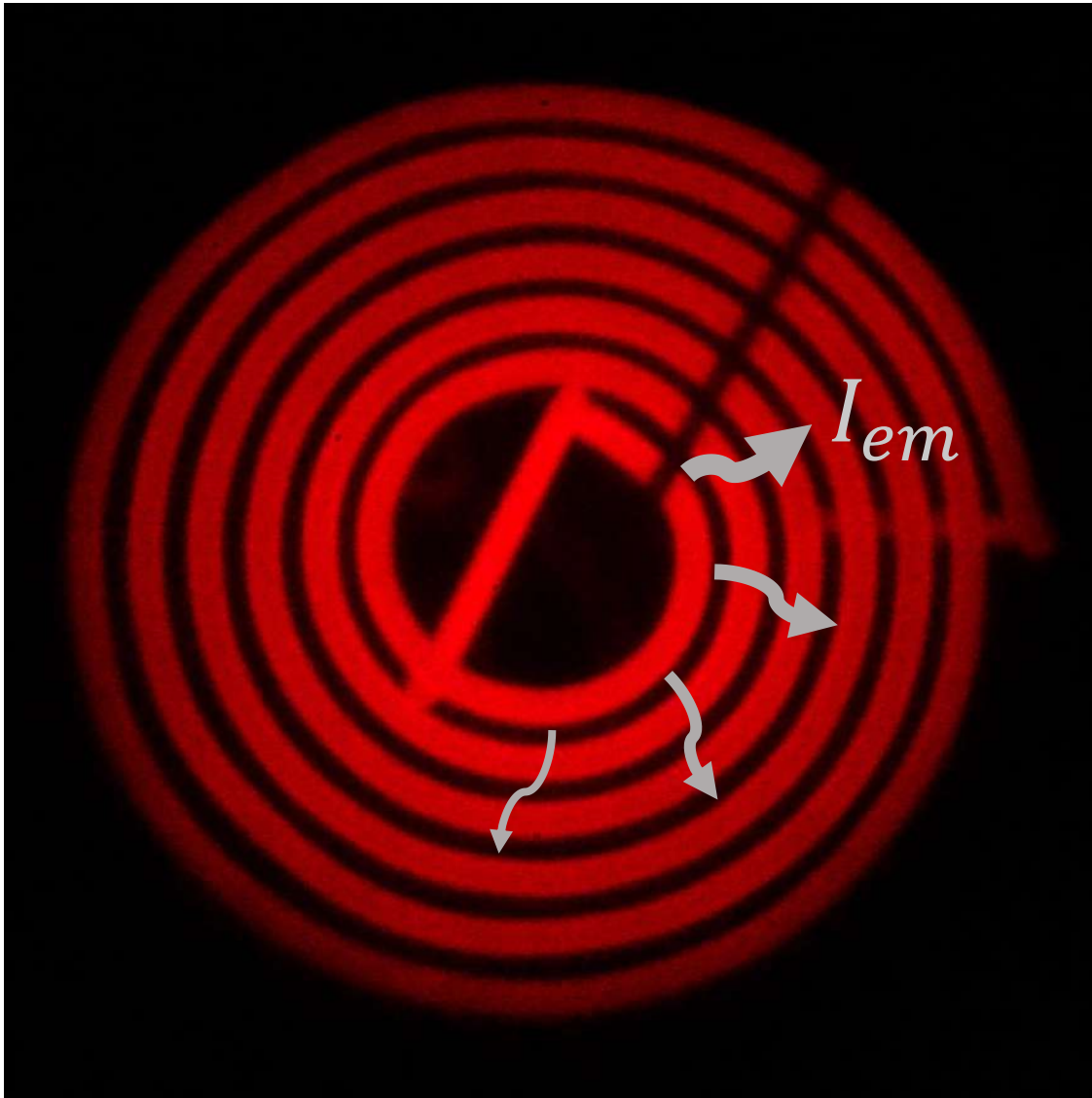
➤ Limits: Refining simulations (electron bombardment...)

Plasma-cathode interactions features



Plasma-cathode interactions features

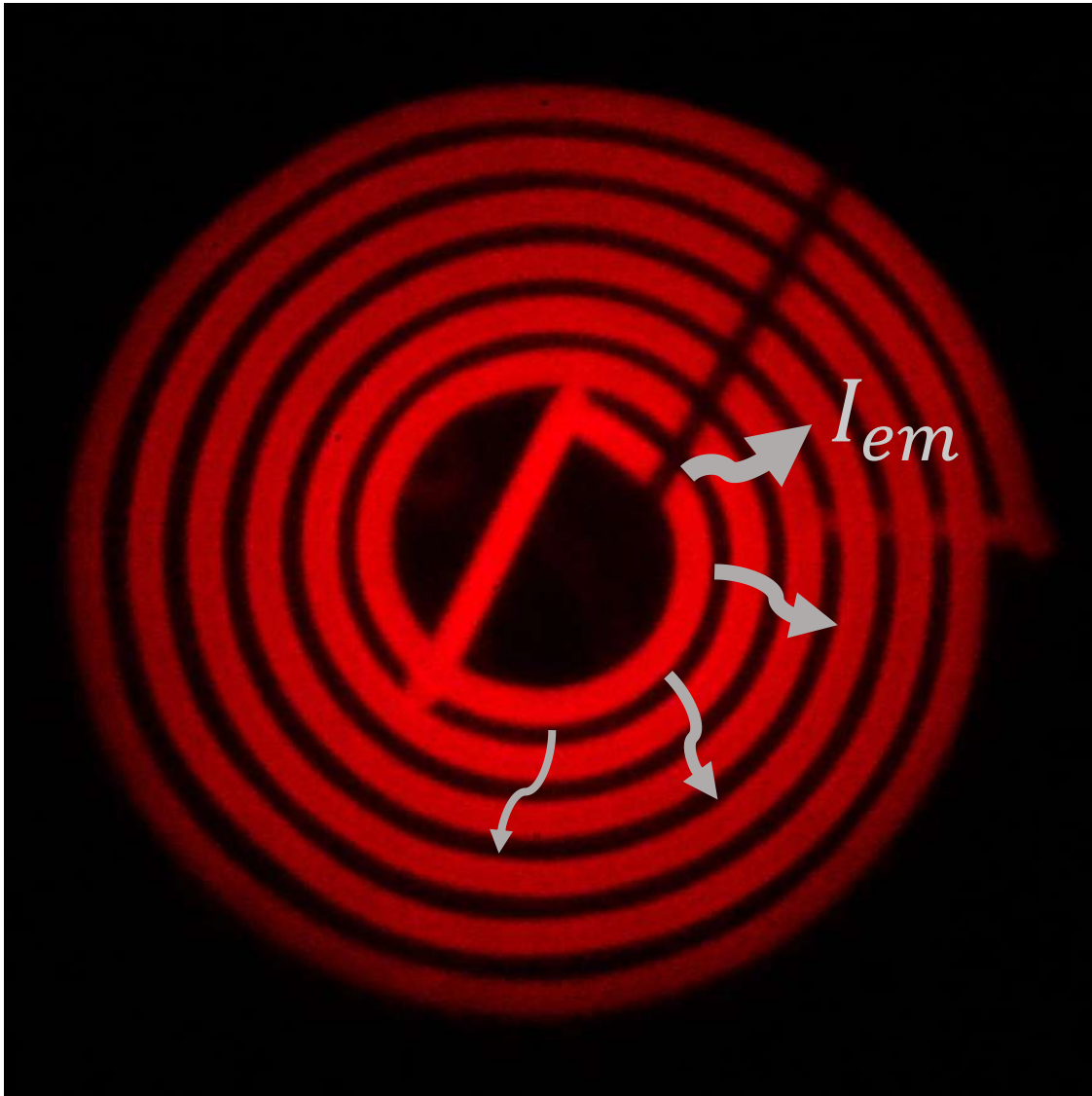




DC heating:

$$\dot{Q}_{\Omega} = R \left(I_h + \int_s^{L_W} i_{em}(u) du \right)^2$$

- Heterogeneous T_W : ohmic heating
- V_b runaway: ion bombardment



DC heating:

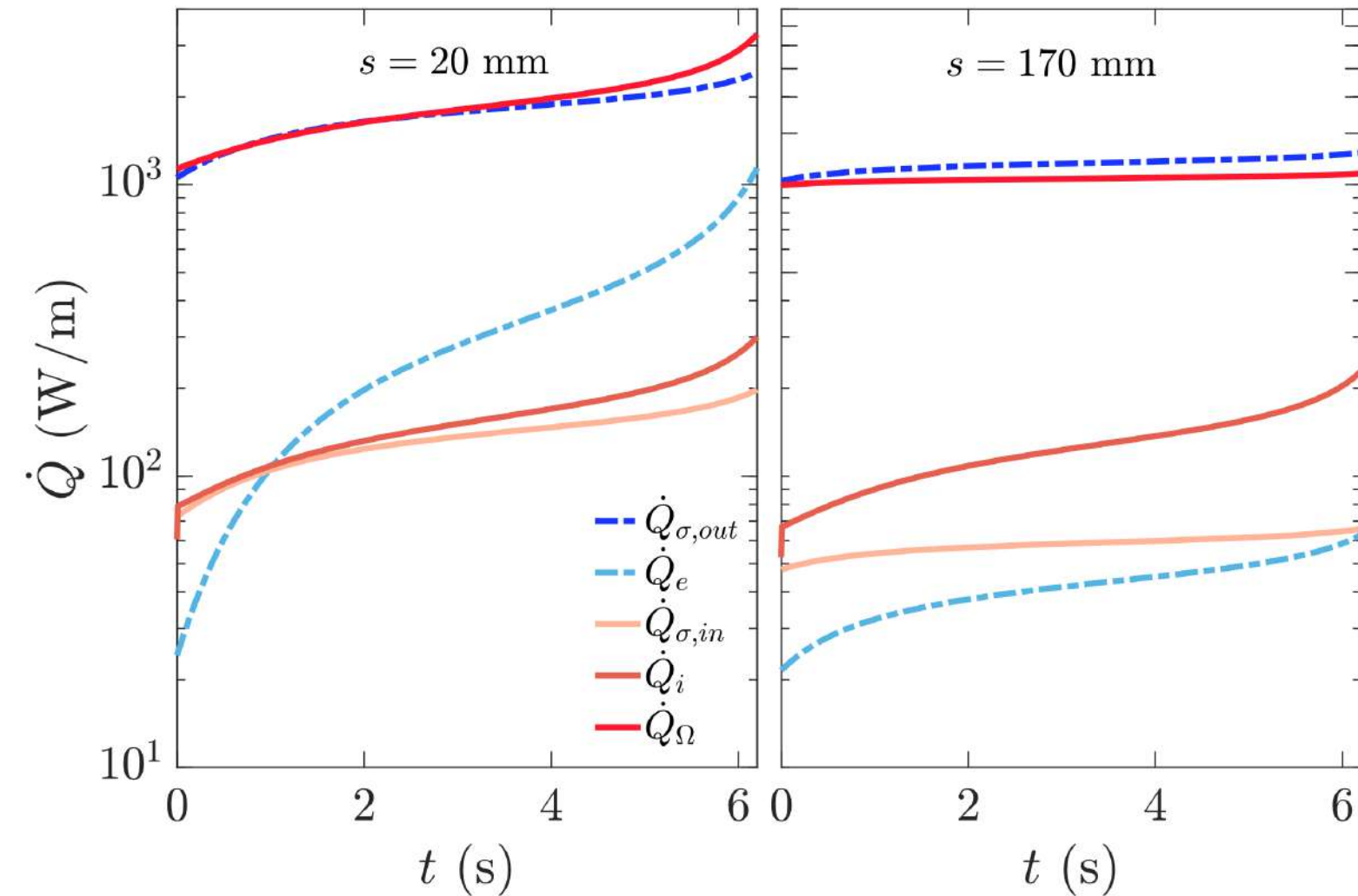
$$\dot{Q}_{\Omega} = R \left(I_h + \int_s^{L_W} i_{em}(u) du \right)^2$$

- Heterogeneous T_W : ohmic heating
- V_b runaway: ion bombardment

Ion bombardment:

$$\dot{Q}_i = \frac{I_i}{e} e [E_i - W + \beta(\phi_p - V_b)]$$

Heat transfers mechanisms



- Initial heating: homogeneous ion bombardment
- Unstable system at $s \rightarrow 0$ mm (ohmic heating)

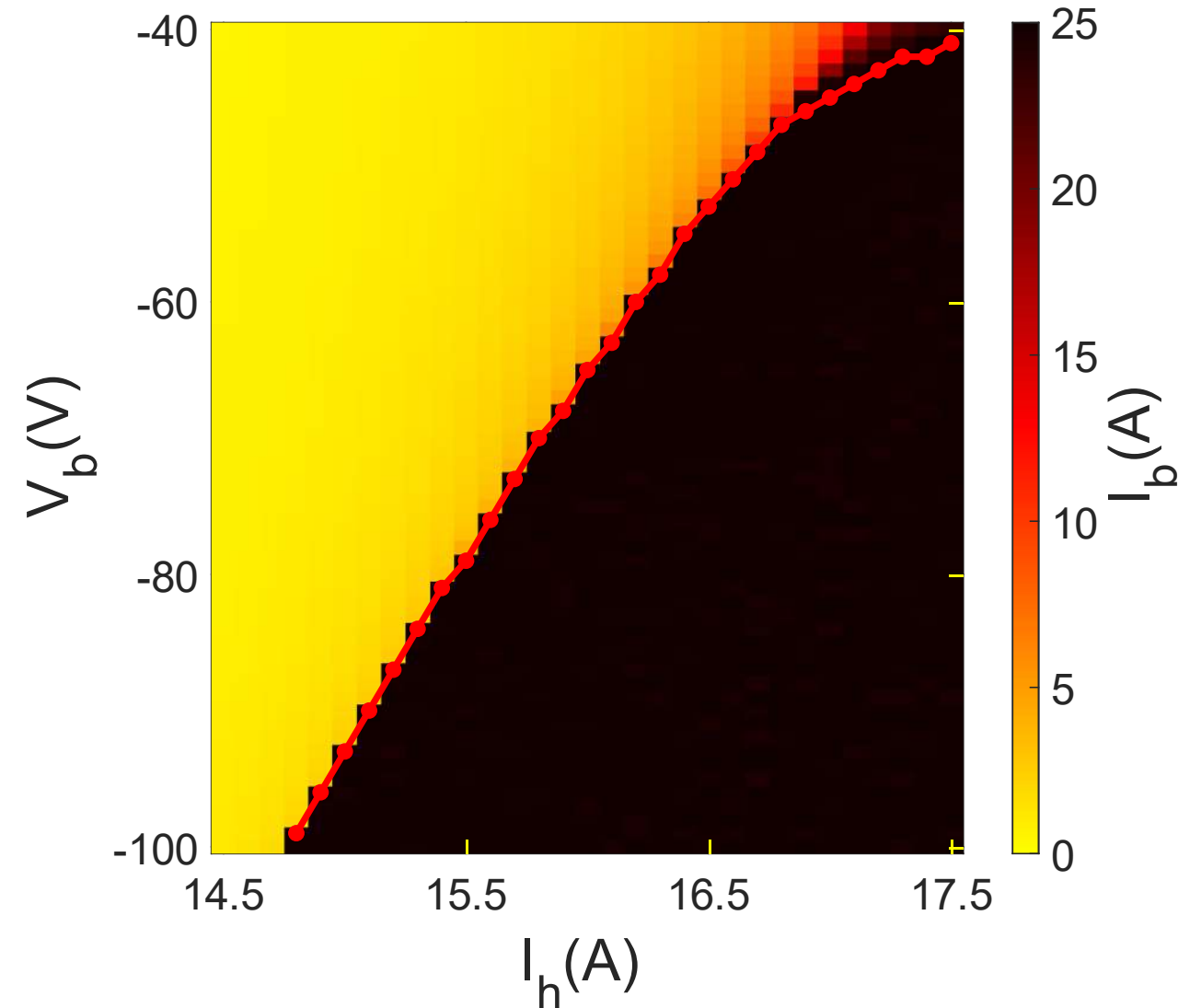


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Applications

I_b prediction and the stability domain

43



- 30 s simulations in the $(I_h; V_b)$ plan
- *Runaway*: reaching $I_b = 25$ A
- Brutal divergence, risk to damage the cathode.

Computing I_{em} experimentally

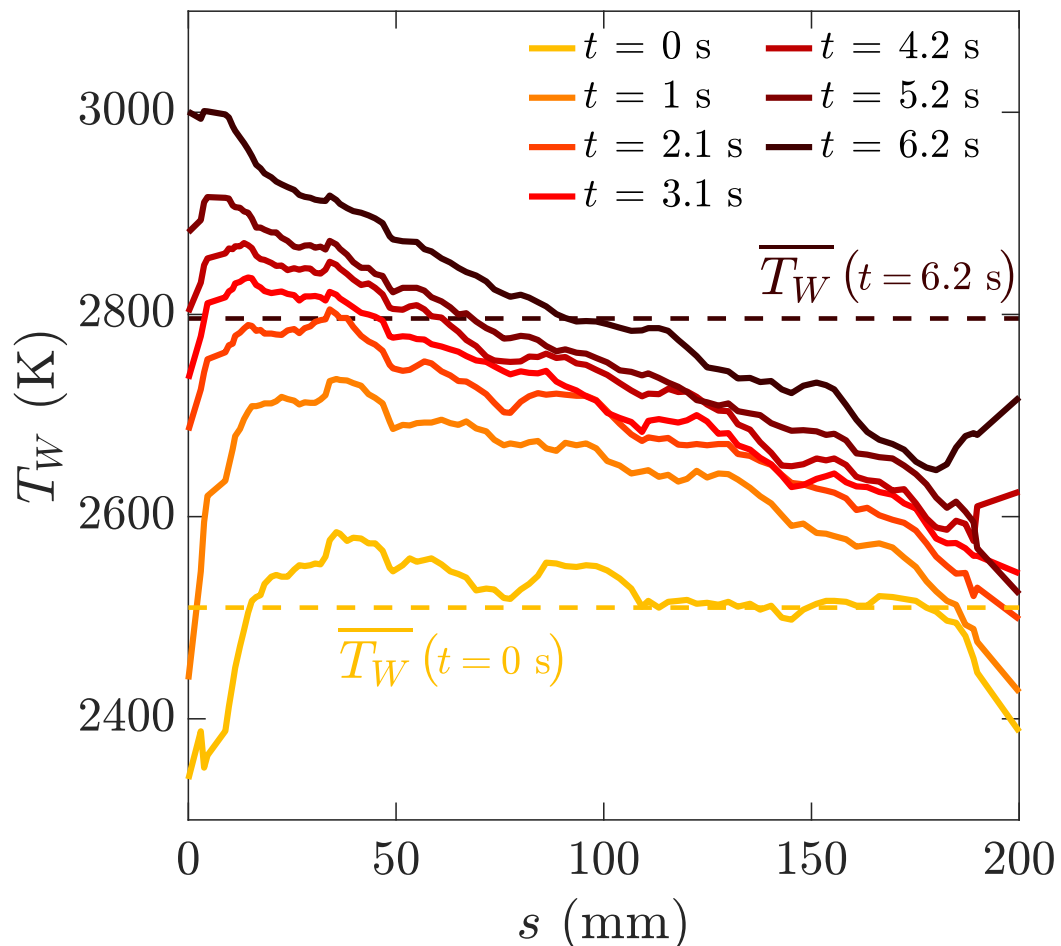
$$\rho_W = a T_W^{1.22} + b$$

Assumption: homogeneous effective $\overline{T_W}$

Computing I_{em} experimentally

$$\rho_W = aT_W^{1.22} + b$$

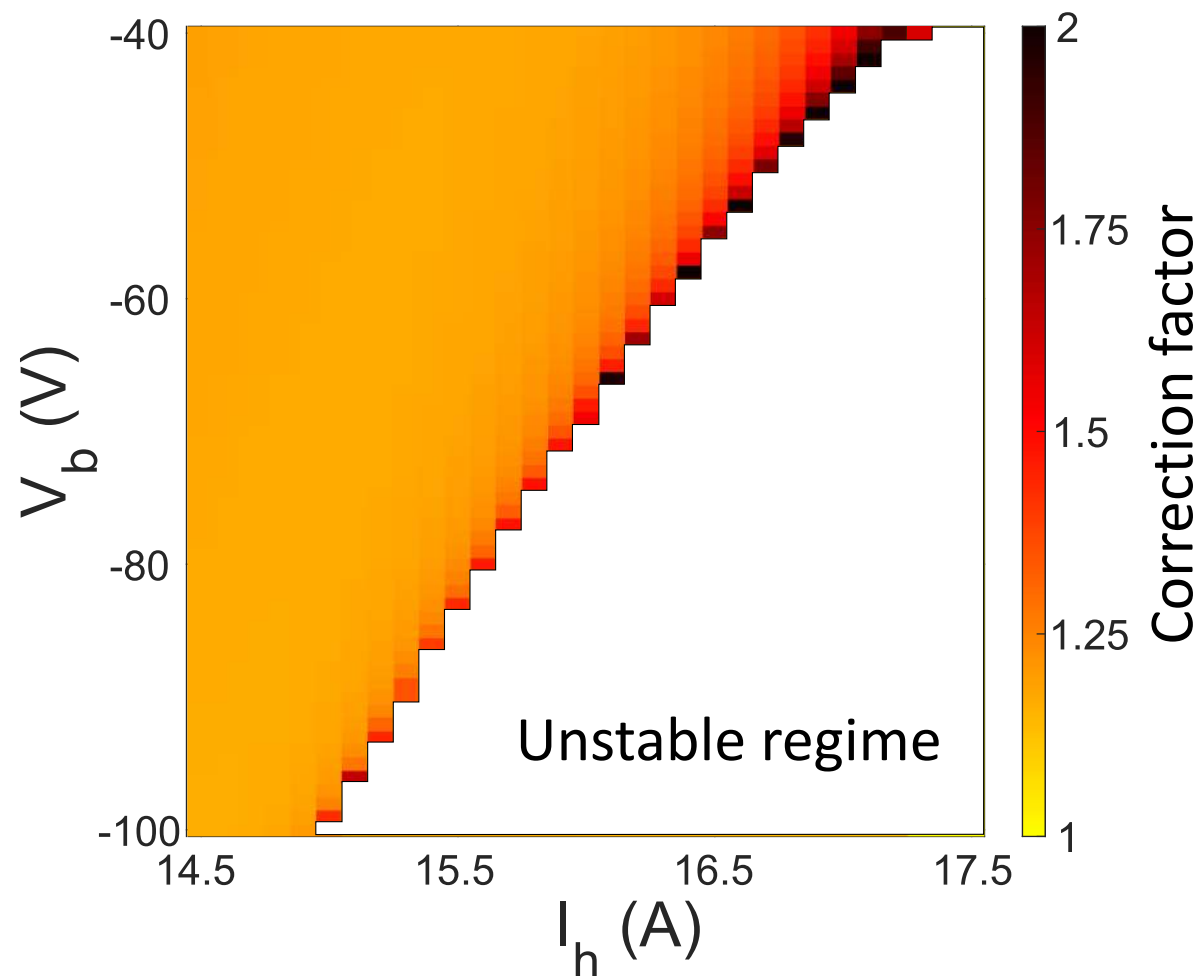
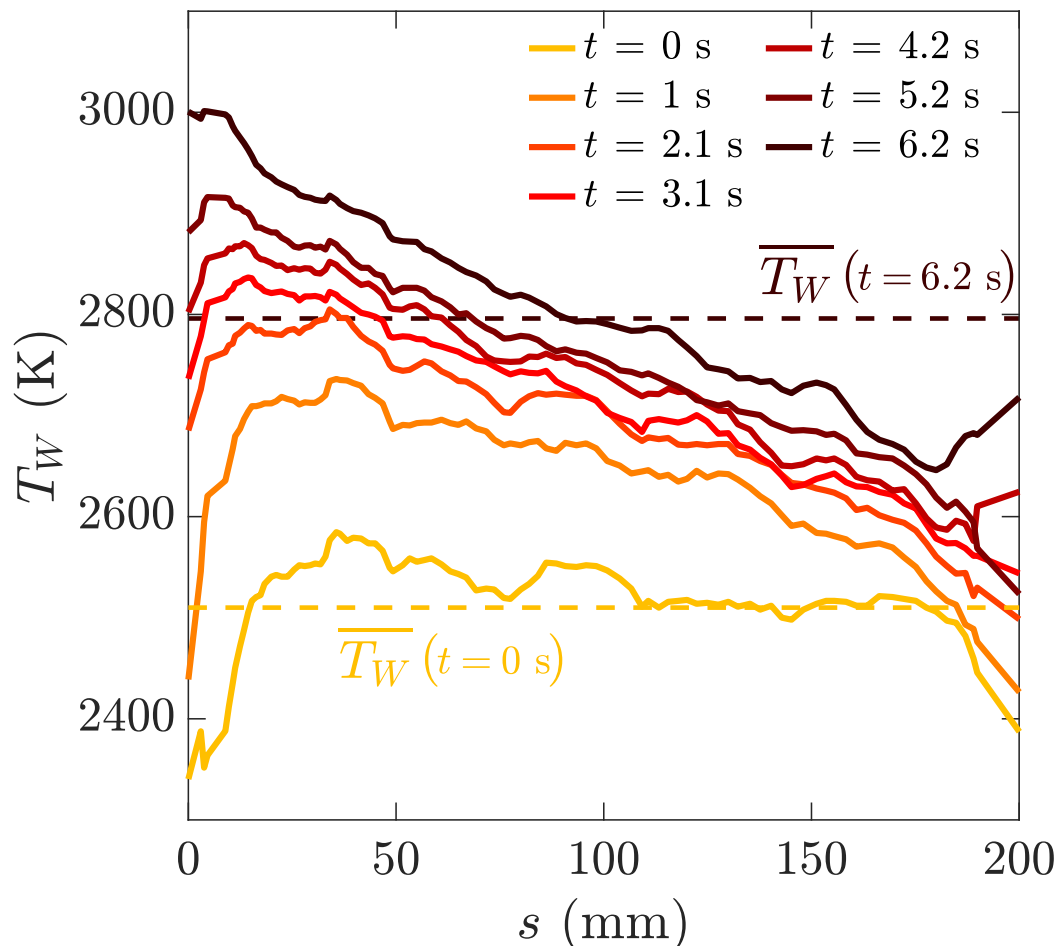
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Computing I_{em} experimentally

$$\rho_W = aT_W^{1.22} + b$$

Assumption: homogeneous effective $\overline{T_W}$



➤ Obtaining I_{em} with a simple resistance measurement

CONCLUSION

- Understanding the pure tungsten cathode
 - Guide for a simple and versatile method
 - Exciting prospects on plasma parameters control
- Modelling the rich plasma-cathode dynamic system
 - Excellent agreement between experiments and model
 - Reliable reproduction of stable and divergent regimes

LTDP poster session on Thursday

Article soon submitted to PSST

Destabilizing effect

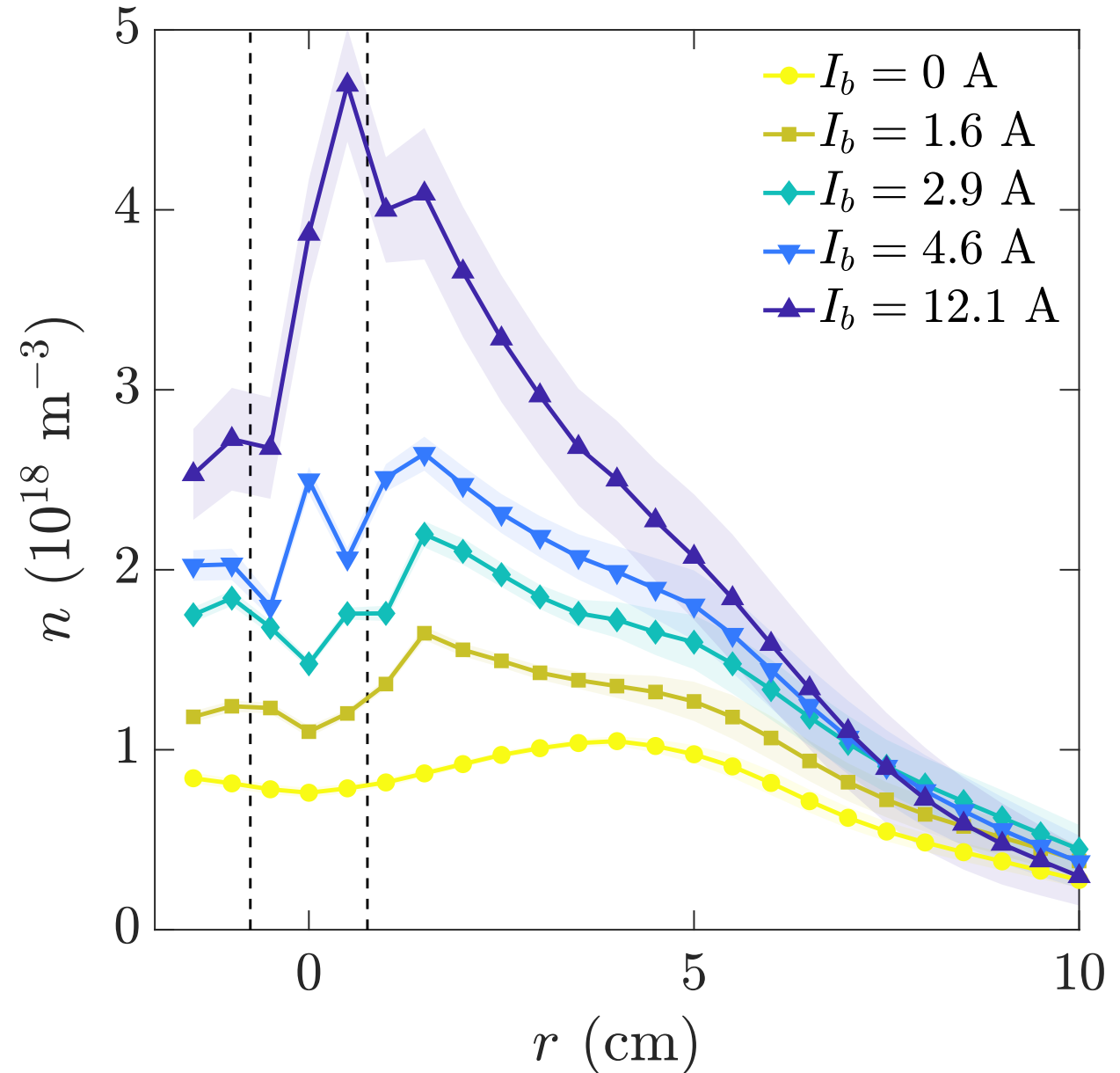
- **Order 0** : New ohmic heating I_b
- **Order 1** : Denser plasma + ion bombardment
- **Order 2** : Rise in T_e

Stabilizing effect

- **Order 0** : Thermionic cooling
Dissipative radiations
- **Order 1** : V_p lowering
- **Order 2** : Space-charge limited effects

Density increase

Thermionic emission causes ionization and enhances the density.



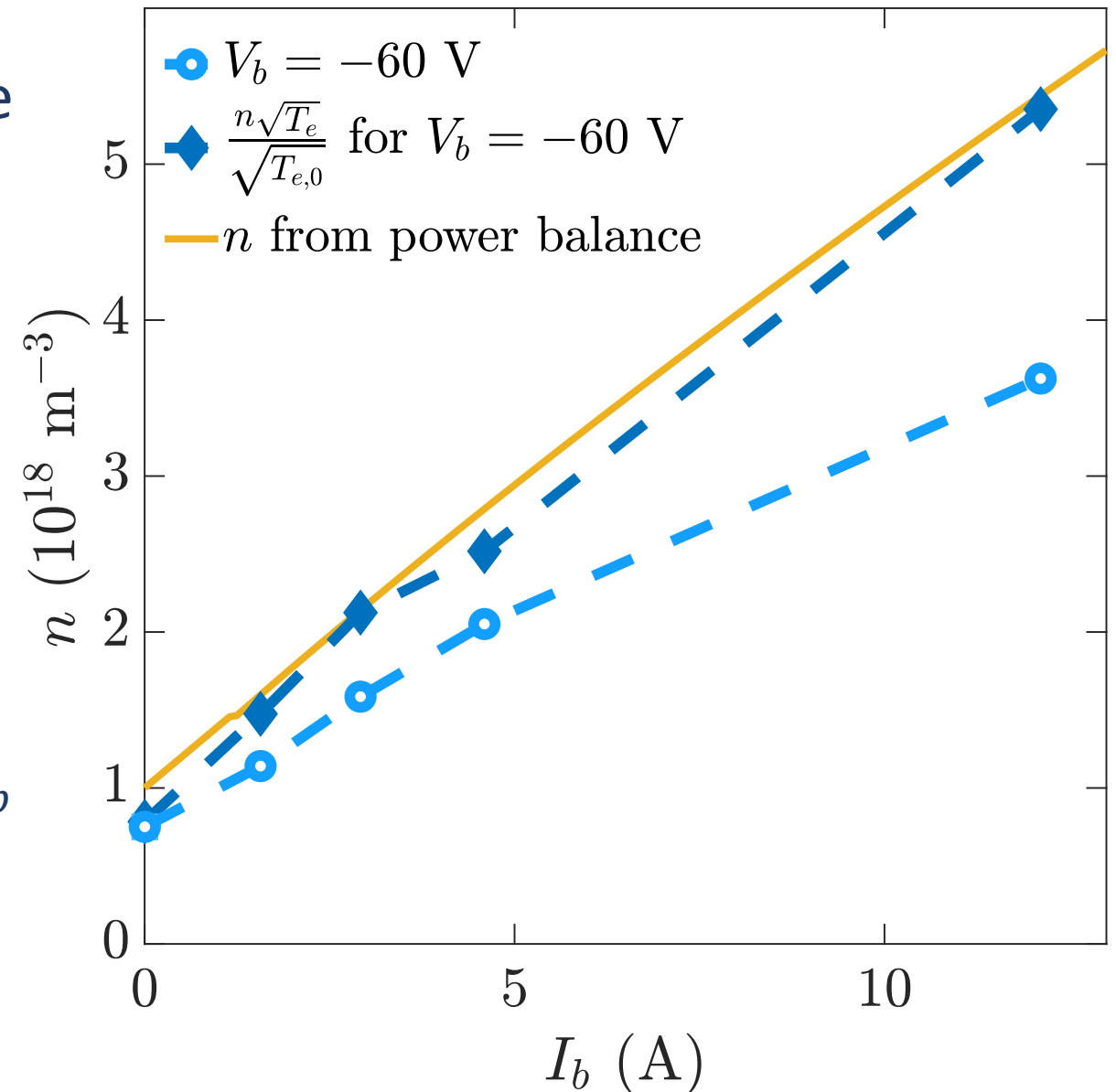
Density increase

50

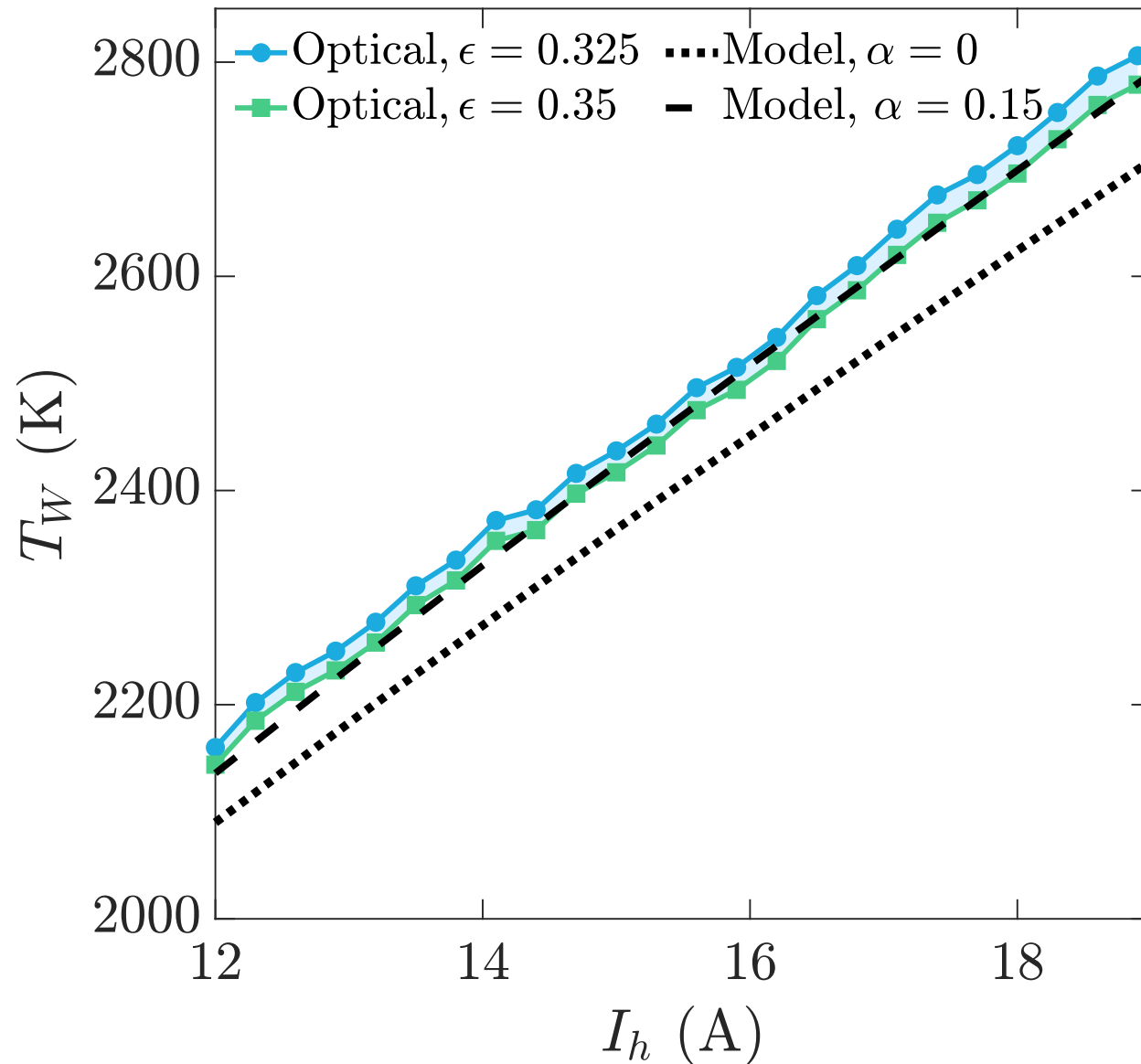
Density computed via a power balance
 T_e considered constant numerically

$$P_{losses} = P_{RF} + P_{cath}$$

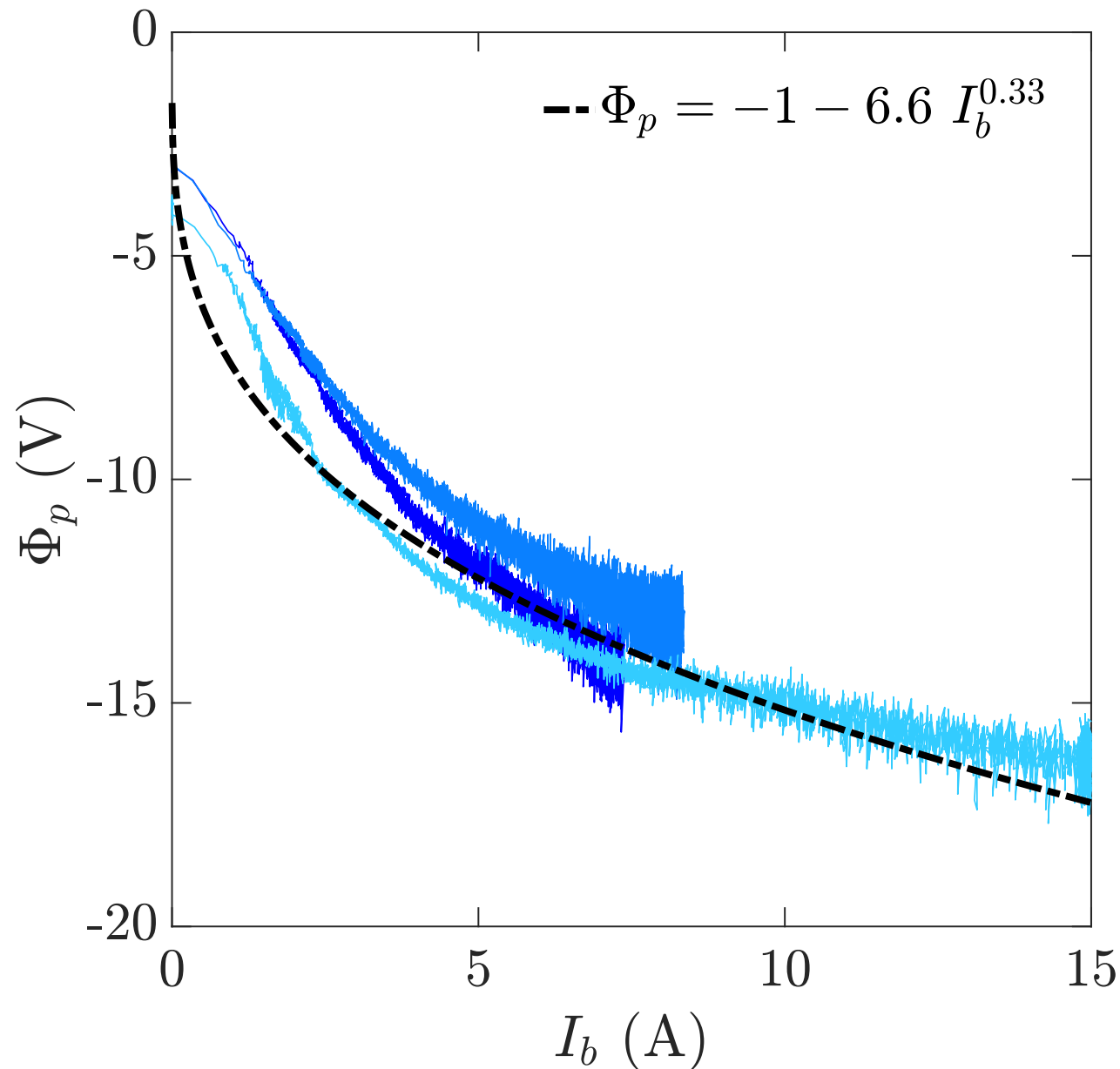
$\nearrow$ Ions-electrons at the walls
 $\nearrow$ 1 kW
 $\nearrow$ Thermionic emission
 Sheath $\Delta V = \Phi_p - V_b$



Setting α

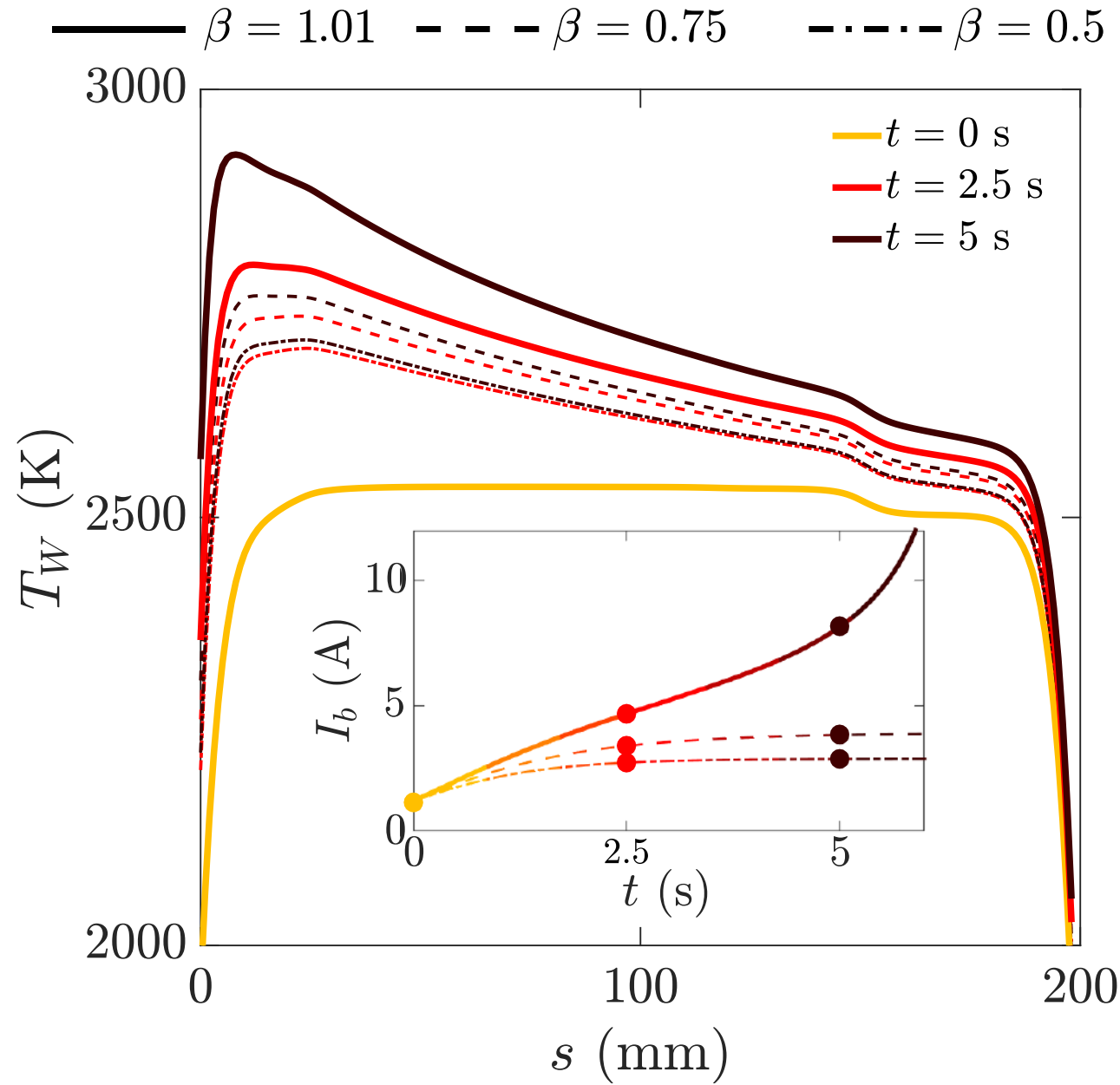


α is set by matching experimental and numerical T_W in absence of plasma



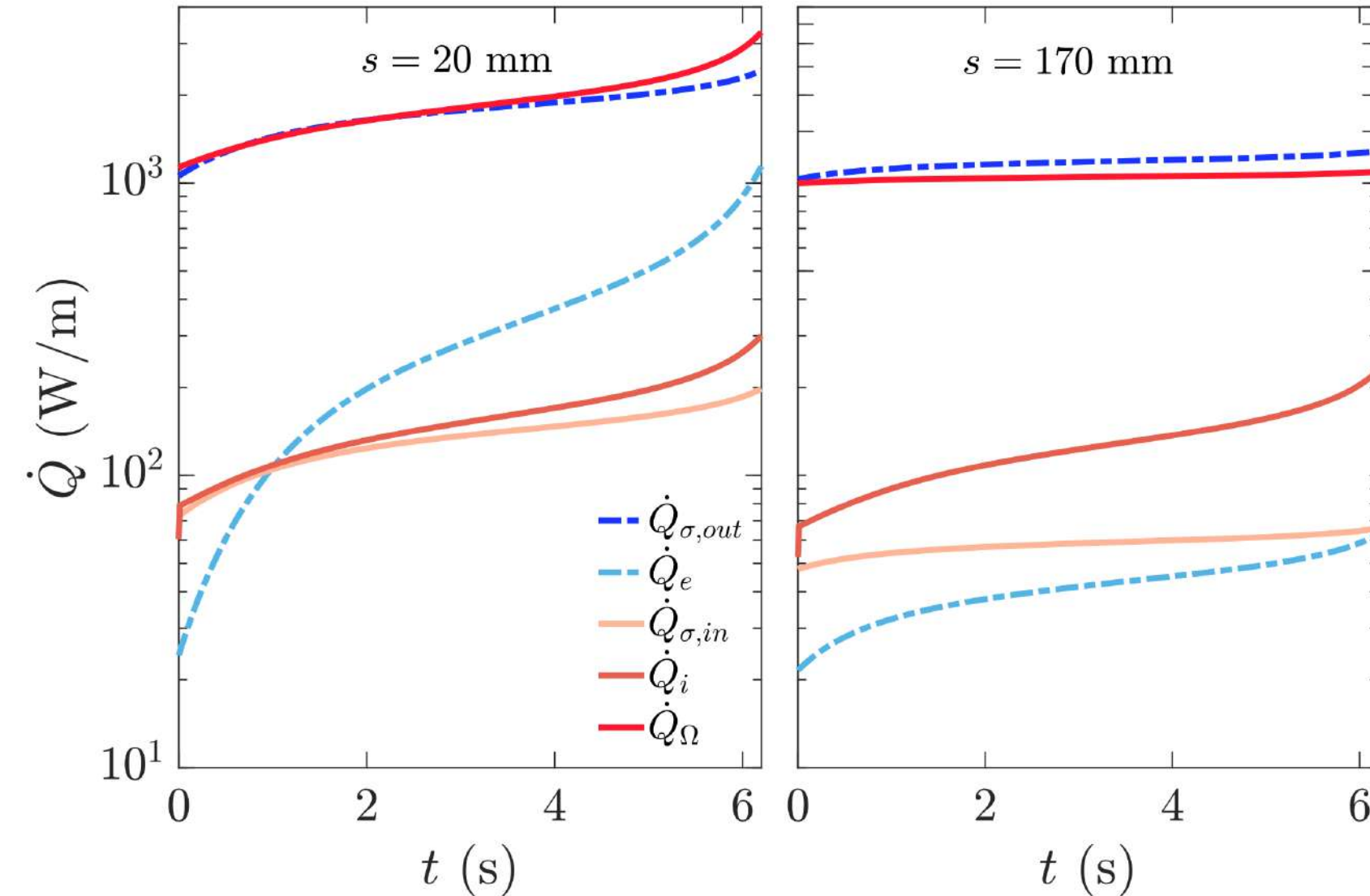
Variations in ϕ_p are
 implemented empirically
 in simulations.

Influence of β



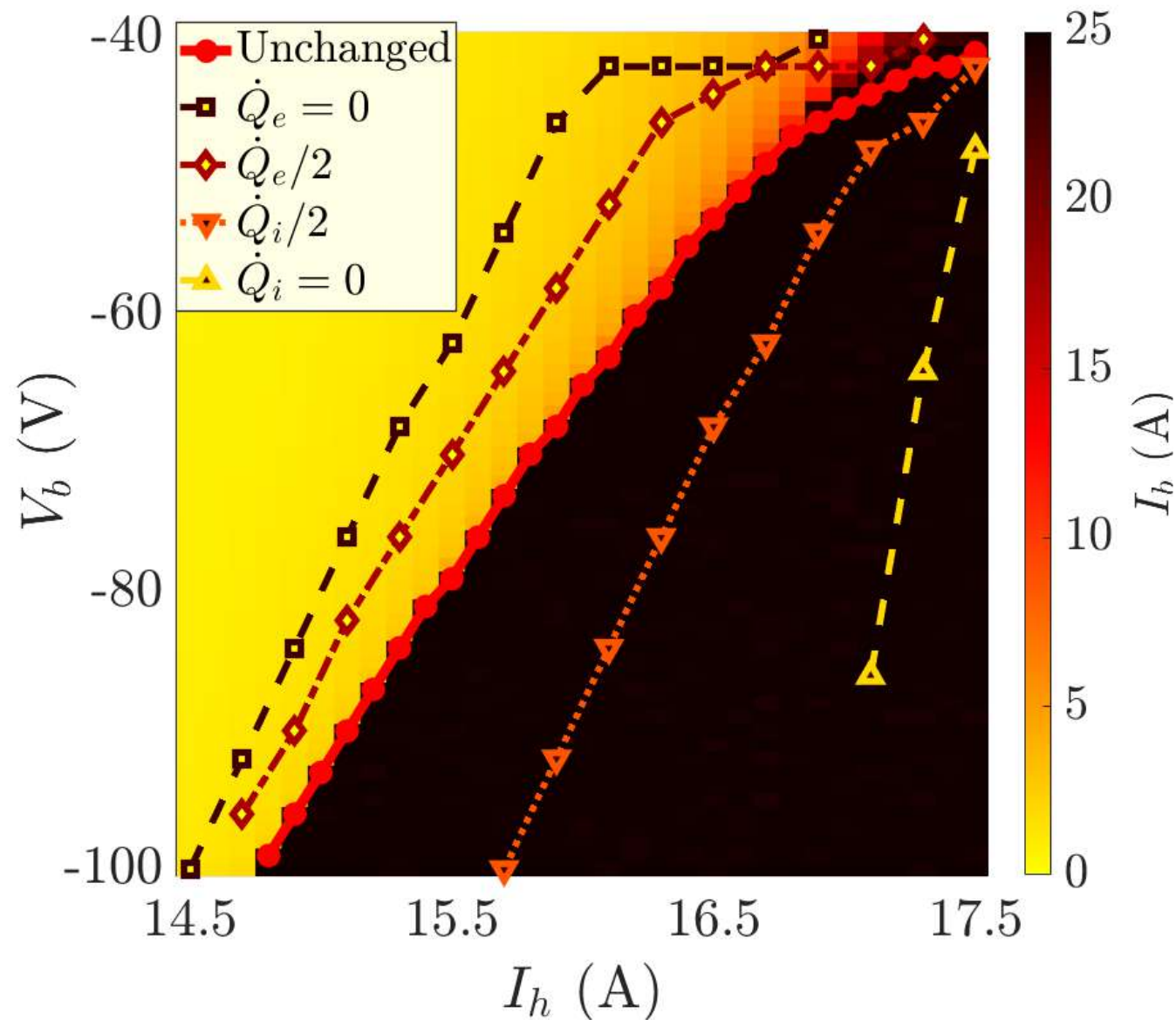
Ion bombardment provides a uniform heating.
The β value is highly influent.

Heat transfers mechanisms

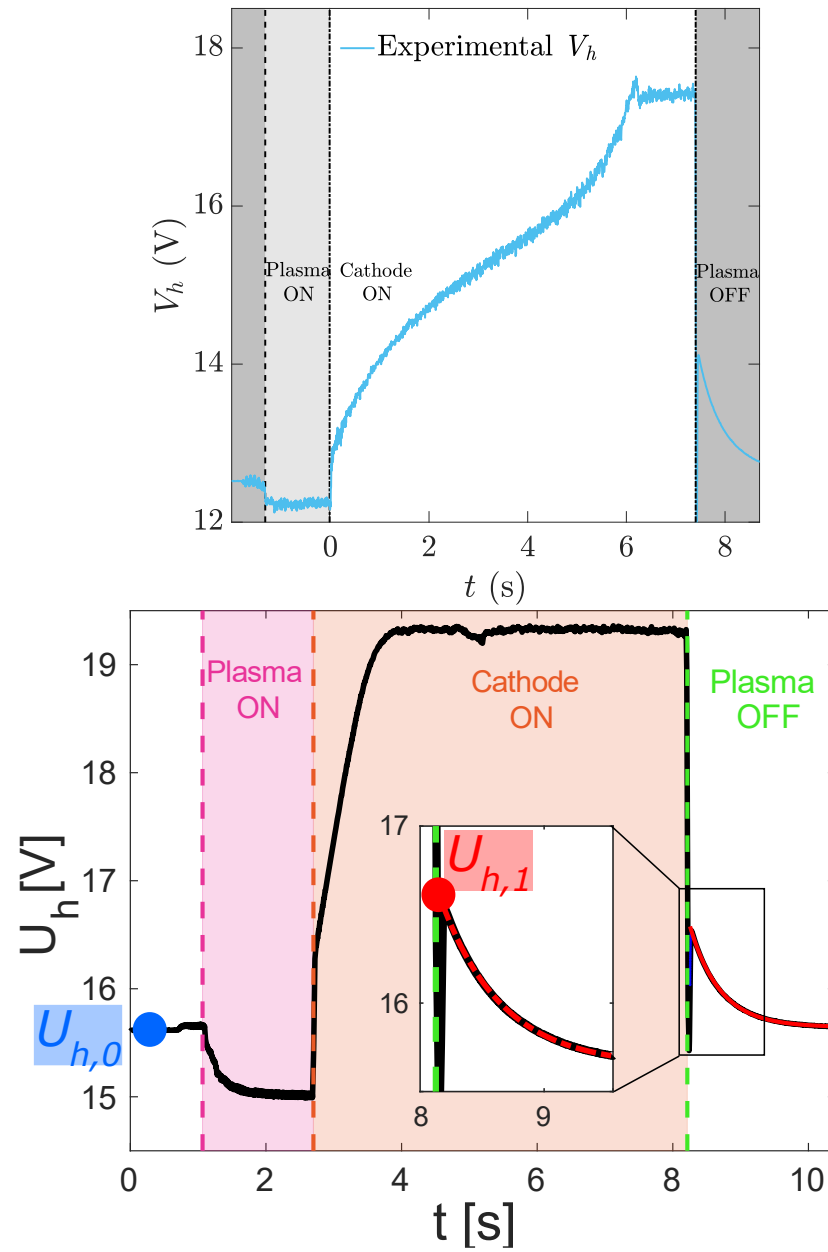


- Unstable system at $s \rightarrow 0$ mm
- Fast increase in ohmic heating
- Homogeneous ion bombardment

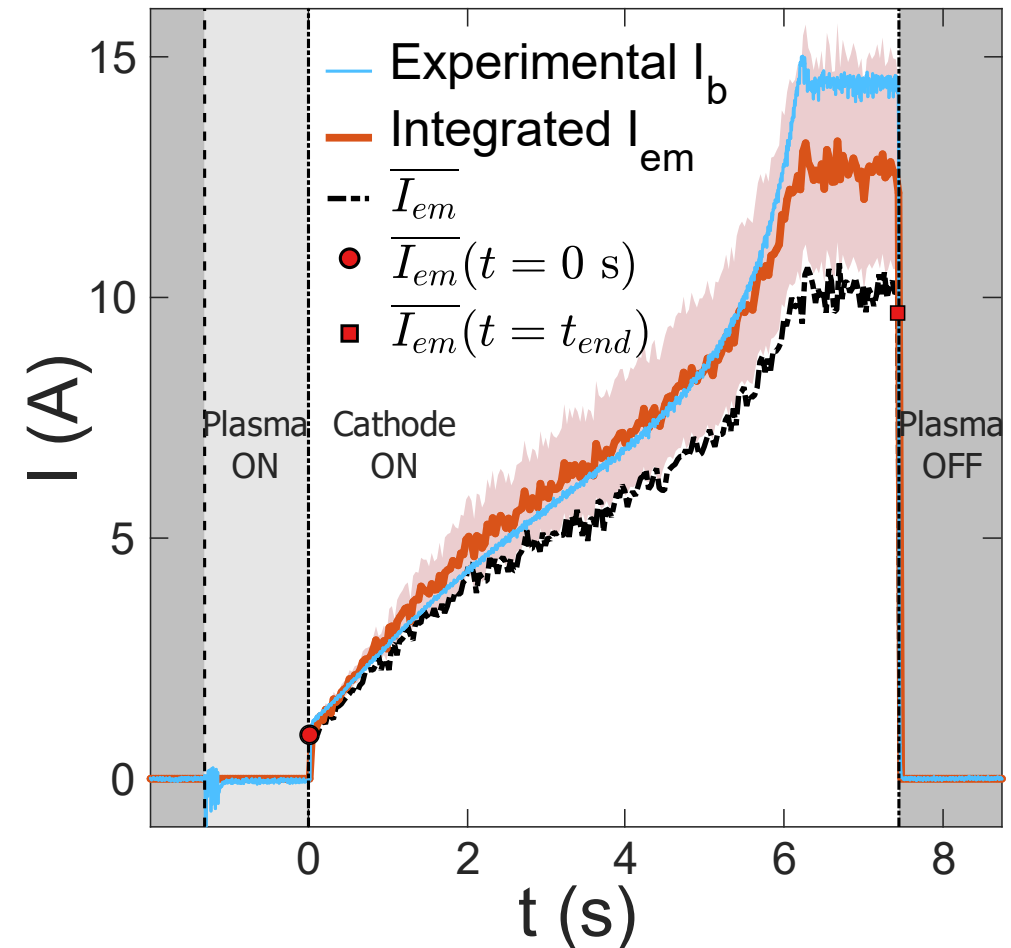
Influence of $\dot{Q}_i$ and $\dot{Q}_e$



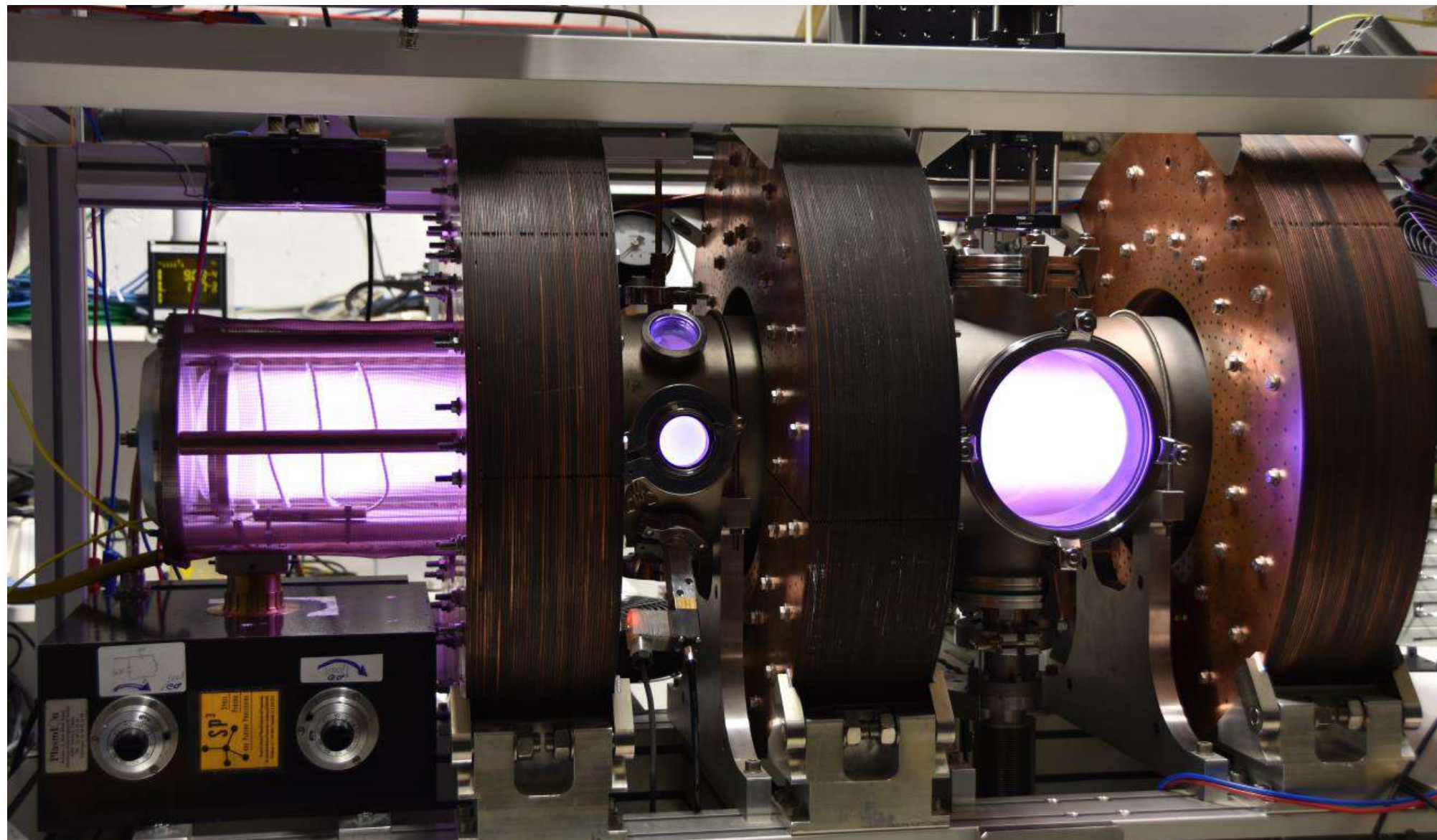
Correcting the resistivity method



$$\rho_W = aT_W^{1.22} + b$$



Experimental setup: VKP



Simulation of an unstable regime

